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Multi-sensor integrated navigation/positioning systems using data fusion: From analytics-based to learning-based approaches☆

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ABSTRACT

Navigation/positioning systems have become critical to many applications, such as autonomous driving, Internet of Things (IoT), Unmanned Aerial Vehicle (UAV), and smart cities. However, it is difficult to provide a robust, accurate, and seamless solution with single navigation/positioning technology. For example, the Global Navigation Satellite System (GNSS) cannot perform satisfactorily indoors; consequently, multi-sensor integrated systems provide the solution, as they compensate for the limitations of single technology by using the complementary characteristics of different sensors. This article describes a thorough investigation into multi-sensor data fusion, which over the last ten years has been used for integrated positioning/navigation systems. In this article, different navigation/positioning systems are classified and elaborated upon from three aspects: (1) sources, (2) algorithms and architectures, and (3) scenarios, which we further divide into two categories: (i) analytics-based fusion and (ii) learning-based fusion. For analytics-based fusion, we discuss the Kalman filter and its variants, graph optimization methods, and integrated schemes. For learning-based fusion, several supervised, unsupervised, reinforcement learning, and deep learning techniques are illustrated in multi-sensor integrated positioning/navigation systems. Design consideration of these integrated systems is discussed in detail from several aspects and their application scenarios are categorized. Finally, future directions for their research and implementation are discussed.

1. Introduction

In recent decades, navigation/positioning systems have become essential parts of many applications, such as driverless cars, the Internet of Things (IoT), Unmanned Aerial Vehicle (UAV), and smart cities. The state-of-the-art navigation/positioning systems include Inertial Navigation [1], Global Navigation Satellite System (GNSS) [2], including Global Positioning System (GPS), BeiDou Navigation Satellite System (BDS), GLObal NAvigation Satellite System (GLONASS), and Galileo, Visible Light Positioning (VLP) [3], WiFi [4], Bluetooth [5], Radio-Frequency Identification (RFID) [6], Ultra-Wideband (UWB) [7], Ultrasonic [8], Magnetic [9], Odometer [10], Vision [11], LiDAR [12], and 5G [13]. Different kinds of sensors, such as inertial sensors, GNSS receivers, photodiodes, wireless receivers, magnetometers, and cameras, are used in these navigation/positioning systems.

It is difficult to use single-sensor-based navigation/positioning systems to provide robust, accurate, and seamless solution, due to their limitations. For instance, the Inertial Navigation System (INS) is a *relative positioning* technology and only provides an accurate solution

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Glossary

ADM	Aircraft Dynamic Model
ANN	Artificial Neural Network
ANS	Autonomous Navigation System
AOA	Angle of Arrival
AP	Access Point
ATEKF	Adaptive Two-stage EKF
AUV	Autonomous Underwater Vehicle
BDS	BeiDou Navigation Satellite System
BLE	Bluetooth Low Energy
CDKF	Central Difference Kalman Filter
CNN	Convolutional Neural Network
CSAC	Chip Scale Atomic Clock
CSI	Channel State Information
DLL	Delay Lock Loops
DNN	Deep Neural Networks
DSRC	Dedicated Short-Range Communication
DVL	Doppler Velocity Log
ECEF	Earth-Centered, Earth-Fixed
EKF	Extended Kalman Filter
EM	Expectation–Maximization
EnKF	Ensemble Kalman Filter
FLAS	Fuzzy Logic Adaptive System
FKF	Federated Kalman Filter
FLL	Frequency Lock Loops
FOV	Field of View
FPLL	Frequency-assisted Phase Locked Loop
GNN	Grey Neural Networks
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
HAPS	High Attitude Platform Station
IF	Intermediate Frequency
ISM	Industrial Scientific Medical
ISRUKF	Iterated Square Root UKF
IMM	Interacting Multiple Module
IMU	Inertial Measurement Unit
IoT	Internet of Things
INS	Inertial Navigation System
JDL	Joint Directors of the Laboratories
KF	Kalman Filter
kNN	k-Nearest Neighbor
LCI	Loosely-Coupled Integration
LED	Light-Emitting Diode
LiDAR	Light Detection and Ranging
LIO	LiDAR-Inertial Odometry
LS-SVM	Least Squares Support Vector Machine
LSTM	Long Short Term Memory
MAP	Maximum A Posteriori
MEMS	Micro Electromechanical Systems
MIMO	Multiple-Input-Multiple-Output
NCO	Numerically-Controlled Oscillator
NLOS	Non-Line-of-Sight
PCA	Principal Component Analysis
PDR	Pedestrian Dead Reckoning
PF	Particle Filter
PLL	Phase Lock Loops
PnP	Perspective-n-Point
POS	Position and Orientation System
PRN	Pseudo Random Noise
PSOBPNN	Particle Swarm Optimization Back Propagation Neural Network
PVA	Position, Velocity, and Attitude
PVT	Position, Velocity, and Time
RANSAC	Random Sample Consensus
RF	Radio Frequency
RFID	Radio-Frequency Identification
RFR	Random forest regression
RISS	Reduced Inertial Sensor System
RMSE	Root Mean Square Error
RNN	Recurrent Neural Network
RSS	Received Signal Strength
RTK	Real-Time Kinematic
SAE	Stacked Auto-Encoder
SAR	Synthetic Aperture Radar
SfM	Structure from Motion
SIG	Special Interest Group
SINS	Strapdown Inertial Navigation System
SLAM	Simultaneous Localization and Mapping

for a limited time, as both inertial sensor errors and integration errors will cause the solution to diverge. Thus, other *absolute positioning* data sources, such as GNSS, are usually needed; however, although they are accurate in open-sky environments, they suffer from signal blockage and multipath in urban areas and other GNSS-challenging environments. Other wireless positioning systems, such as WiFi and Bluetooth, usually have some limitations, such as high dependency on the distribution of Access Points (APs), noisy and unstable solutions, labor costs for building databases, and the fluctuation of Received Signal Strengths (RSS) indoors [14]. Magnetic positioning is usually used indoors for local, as opposed to global positioning. Vision positioning, which often uses a camera to capture an object's motion, offers accurate localization at a relatively low cost [15]; however, it has limitations, such as privacy issues, the extraction of features from environments, and large computation load. It is difficult, therefore, to achieve navigation/positioning goals while satisfying application requirements, by using a single-sensor modality.

Consequently, to improve positioning performance, the advantages of different kinds of sensors can be fully exploited to improve reliability, robustness, and spatial and temporal coverage. Taking a GNSS/INS integrated system as an example, GNSS provides the position and velocity to aid inertial navigation by reducing cumulative errors, while INS fills the gap in challenging environments by filtering its noise; hence, integration reduces the limitations of both constituent parts. Many similar positioning systems can be integrated with INS, such as WiFi, Bluetooth, RFID, UWB, Ultrasonic, Magnetic, and Vision, which can be divided into three categories: (i) Loosely-Coupled Integration (LCI), (ii) Tightly-Coupled Integration (TCI), and (iii) Ultra-Tightly-Coupled Integration (UTCI). LCI integrates different sources, based on position and velocity levels, while TCI integrates sensor data in ranging levels between transmitters and the receiver. Different from LCI and TCI, UTCI integrates at a deeper level of raw measurements, such as raw carrier phase and code phase, from a GNSS receiver. Multi-sensor integrated systems can be used in many applications, such as aircraft, underwater vessels, surface ships, spacecraft, cars, indoor mobile robots, and UAVs.

Data fusion, which fuses diverse information from various sources, is the most important part of integrated systems, a widely accepted definition of which was provided by the Joint Directors of the Laboratories (JDL) workshop [16]: "A multi-level process dealing with the

SLFNs	Single-hidden Layer Feed-forward Networks
STKF	Strong Tracking Kalman Filter
STL	Scalar Tracking Loop
SVM	Support Vector Machine
TDOA	Time Difference of Arrival
TOA	Time of Arrival
TCI	Tightly-Coupled Integration
TXCO	Temperature Compensated Crystal Oscillator
UAV	Unmanned Aerial Vehicle
UHF	Ultra-High Frequency
UKF	Unscented Kalman Filter
UPF	Unscented Particle Filter
USBL	Ultrashort Base Line
UTC	Coordinated Universal Time
UTCI	Ultra-Tightly-Coupled Integration
UTP	Underwater Transponder Positioning
UWB	Ultra-Wideband
VINS	Visual-Inertial Navigation System
VIO	Visual-Inertial Odometry
VLC	Visible Light Communication
VLP	Visible Light Positioning
VNS	Visual Navigation System
VO	Visual Odometry
VTL	Vector Tracking Loop
VTL_DI	Vector Tracking Loops Based Deep Integration
WLAN	Wireless Local Area Network
WSN	Wireless Sensor Network
1DWF	One-Dimensional Wiener Filter

association, correlation, combination of data, and information from single and multiple sources to achieve refined position, identify estimates and complete and timely assessments of situations, threats, and their significance". There are several categories of data fusion algorithms; however, analytics-based and learning-based approaches are the most widely used. An analytics-based approach uses an analytics function to model system states and external measurements, which are called 'estimations' in the literature [17]. Many analytics-based methods are widely used, such as Kalman Filter (KF) and Particle Filter (PF), while several learning-based data fusion methods, such as Artificial Neural Network (ANN), Fuzzy Logic, and Support Vector Machine (SVM) are also used, since they can model systems without prior statistical information about the process and measurement noise. Learning-based methods sometimes perform better than analytics-based methods.

The literature relating to data fusion for integrated positioning/navigation systems was investigated in [15,18–21]. Loebis et al. [18] surveyed developments in Autonomous Underwater Vehicles (AUVs) navigation and multi-sensor data fusion techniques to improve the AUV's navigation capability. Smith and Singh [19] covered the applications of various algorithms at different layers of the JDL model, and highlighted the weaknesses and strengths of their applications. Ben-Afia et al. [15] presented a classification of vision-based fusion techniques in unknown environments. Yassin et al. [20] surveyed wireless-based indoor positioning technologies and generally introduced hybrid systems combining inertial sensors, cameras, and map matching. Guo et al. [21] presented fusion-based indoor positioning techniques and systems bearing three fusion characteristics: source, algorithm, and weight spaces.

A comparison of these surveys and our work is shown in Table 1. The article [18] concentrated only on the navigation applications of

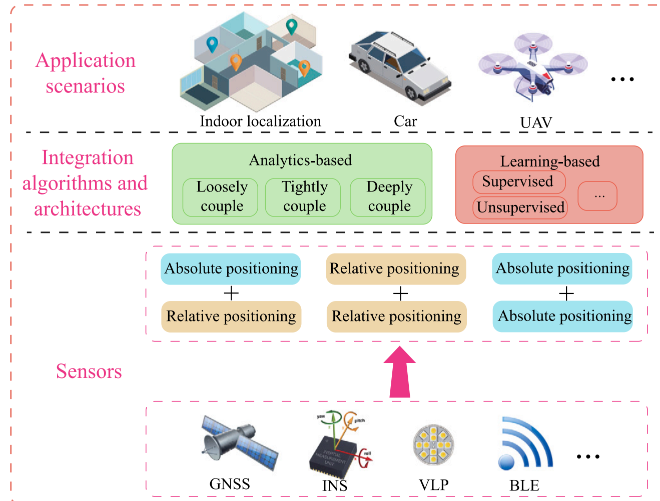

Fig. 1. The overall architecture of the whole system.

underwater vehicles with a limited range of sources. The survey [19] focused on the JDL framework but did not discuss positioning/navigation sources and application scenarios. Moreover, both [18,19] are out of date. The survey [15] only focused on vision-based positioning and did not mention any learning-based methods. The article [20] systematically discussed wireless positioning but only introduced a small range of sensor fusion indoors. The survey [21] discussed both analytics-based and learning-based methods, and gave an in-depth survey on integration sources. But their topic is limited to indoor positioning and their discussions on algorithms and design consideration are not comprehensive. In addition, optimization-based methods and integration scheme-based classifications were not explicitly discussed by either of these surveys.

We organize the text from three layers: sensors, integration algorithms and architectures, as well as application scenarios. The overall architecture of the system is depicted in Fig. 1. The detailed layout of this article is shown in Fig. 2. The main contributions are:

- Based on the sensor integration, we classified multi-sensor fusion into (i) absolute/relative, (ii) relative/relative, and (iii) absolute/absolute integration. Innovatively, we classify absolute positioning sources into five categories: (1) radio-based, (2) light-based, (3) audio-based, (4) field-based, and (5) vision-based, based on their physical properties. The three categories of multi-sensor combinations are also subdivided in our article, which expands the depth of Guo et al. [21].
- The focus here is on analytics-based and learning-based systems, grounded on data fusion algorithms. Most systems are interdisciplinary and have been designed for multiple scenarios. Specialized schemes such as Visual-Inertial Odometry (VIO) are not discussed in detail. In addition, the algorithms and integration architectures of analytics-based systems have been surveyed from three aspects: loosely-coupled, tightly-coupled, and ultra-tightly-coupled.
- We further discuss the analytics-based algorithms and integration architectures. Filtering-based navigation systems have dominated in the past several decades; however, analytics-based algorithms are not limited to filtering-based methods, but also to graph optimization. We also subdivide the learning-based positioning and navigation methods into supervised learning, unsupervised learning, deep learning, and reinforcement learning.
- We classify application scenarios into six categories, each one with a certain number of examples and analyses. Some hot topics, e.g., automated driving and UAVs, are explicitly discussed.

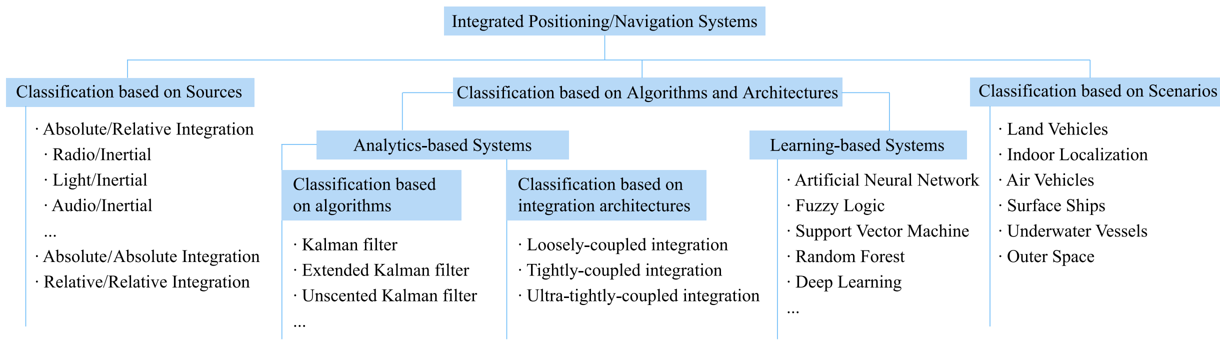

Fig. 2. Overall classification of this article with respect to integrated positioning/navigation systems.

Table 1
Comparison of previous works with ours.

Features	[18]	[19]	[15]	[20]	[21]	Ours
Published year	2002	2006	2014	2017	2020	–
Classification of integrated navigation/positioning systems based on sources	Limited	NO	Limited	Limited	Limited	**In-Depth**
Classification of analytics-based data fusion	General	Limited	General	General	**Explicit**	**Explicit**
Optimization-based algorithms	NO	NO	NO	NO	NO	**Explicit**
Classification of integration architectures	NO	NO	General	NO	NO	**Explicit**
Classification of learning-based data fusion	NO	Limited	NO	General	General	**Explicit**
Design consideration	General	General	NO	General	General	**Explicit**
Application scenarios	Underwater only	NO	NO	Indoor only	Indoor only	**Wide coverage**
Future research directions	General	General	General	**Explicit**	**Explicit**	**Explicit**

- We present a comprehensive tutorial on design consideration. This section discusses the selection between analytics-based and learning-based systems, state selection and observability, real time consideration, scalability, time synchronization, and the robustness of outliers. We also give some future research directions for the field of integrated navigation/positioning.

This article will discuss integrated positioning/navigation systems, thus: Section 2 discusses inertial navigation systems, various wireless positioning systems, and systems that use different sources; Section 3 presents analytics-based multi-sensor data fusion methods, and Section 4, learning-based multi-sensor data fusion systems; Section 5 presents design consideration before a system is implemented; Section 6 presents and categorizes application scenarios of the various systems; Section 7 discusses future directions for research and implementation; Section 8 presents a summary of our work.

2. Positioning/navigation sources

In this section, we present widely-used single positioning/navigation systems and classify integrated positioning/navigation systems concerning source categories.

2.1. Single positioning/navigation systems

Here, we present some widely-used single positioning/navigation systems, including INS, GNSS, Visible light communication, WiFi, Bluetooth, RFID, ultrasonic, magnetics, odometer, vision, LiDAR, and 5G. Some future positioning technologies (e.g., Terahertz-band localization for 6G) have not yet been used, thus they are not covered.

2.1.1. Inertial navigation

Inertial navigation was developed several decades ago for military applications, such as missiles, aeroplanes, and ships, by using inertial sensors, including accelerometers and gyroscopes [1]. Inertial sensors can be navigation grade, tactical grade, auto grade, and low-cost grade [22]; their performance and cost are significantly different. Inertial navigation uses the measurements from accelerometers and gyroscopes to estimate the position, velocity, and attitude of the platform through algorithms, such as INS mechanization, Pedestrian Dead Reckoning (PDR), and motion constraints [23]. Due to errors in inertial sensors and integral computations, as used in inertial navigation algorithms, their accuracy degrades accumulatively, depending on inertial sensor errors [24]. By using self-contained inertial sensors, inertial navigation is independent of any external information, so it is not affected by external electromagnetic interference. Inertial navigation can be used in many environments – sky, ground, and underwater – and applied to both vehicles and pedestrians. It offers position, velocity, and attitude solutions with a high update rate, short-term accuracy, and good stability; its drawback, though, is that navigational errors increase over time, thus its long-term accuracy is poor [25].

2.1.2. Global navigation satellite system

GNSS was developed in the 1960s to provide a real-time, all-weather, global, land/sea/air navigation service in intelligence gathering, nuclear explosion monitoring, and emergency communications [2], since it can measure the distance between satellite and receiver by using Time of Arrival (TOA) combined with data from multiple satellites, to estimate a location to an accuracy ranging from a few meters to centimeters [26]. However, since GNSS is vulnerable to interference from signal blockage, multipath, and electromagnetism [27], it is not accurate in either urban canyons or indoors. To overcome these disadvantages, it must be integrated with other positioning technologies, such as INS [26,28–32], WiFi [33], and UWB [29,34–36]. INS is an ideal complementary data source since integration reduces the disadvantages of inertial dead-reckoning and GNSS's line of sight. When a GNSS signal is interrupted, or blocked, the inertial sensor enables the system to coast until it is re-established. The integration of GNSS and INS improves the quality and integrity of each module and also allows calibration of inertial instrument biases, while the inertial sensors improve the performance of the GNSS receiver [37].

2.1.3. Visible light positioning

Visible Light Communication (VLC), which uses optical light instead of radio waves to transmit data [38], is a relatively recent development that has many advantages over other systems, such as high energy efficiency, long lifetime, low heating, high data rate; it does not harm the human body [39,40]. Consequently, it is now applied to positioning in the navigation field and realizes a new positioning technology called VLP [3]. A practical VLP system usually includes Light-Emitting Diode (LED) lamps as the transmitters, and a photodiode, or camera, as the receiver. The optical signals can be interpreted in many modes [41], such as RSS, TOA, Time Difference of Arrival (TDOA), and Angle of Arrival (AOA). Recent studies have applied several positioning algorithms, such as proximity [42], trilateration [43], multilateration [44], and fingerprinting [45], for each mode. Accuracy of most VLP systems is a few centimeters because the optical signal is not sensitive to fading, severe multipath interference, and electromagnetic wave interference [3]. However, VLP is affected by shadowing or blocking as the optical signals cannot pass through opaque objects.

2.1.4. WiFi

Currently, WiFi is used almost everywhere, from IoT to Smart Cities, which support several IEEE 802.11 protocols; it has been researched for around twenty years [4] and its hardware usually includes WiFi routers or APs as the transmitters and WiFi receivers as receivers. Flight times, arrival angle, Channel State Information (CSI), and RSS are common measurements in WiFi positioning systems. Many algorithms, such as trilateration [4], fingerprinting [4], and Kalman filtering [46], are commonly used in them. WiFi-based positioning systems have the advantages of supporting smart devices, acceptable positioning accuracy (1–10 m), and good portability; also, they do not require additional infrastructure, and have both wide coverage and low cost [47]. However, WiFi signals are susceptible to interference from multipath, signal blockages, AP distribution, measurement fluctuation [23], and extensive labor costs for database building [48]. As for algorithms of data fusion, analytics-based methods, such as Extended Kalman Filter (EKF) [23] and optimization methods [49] have been used in the WiFi/inertial system for indoor pedestrian and robot applications.

2.1.5. Bluetooth

Bluetooth is a short-range data exchanging technology that uses short-wavelength Ultra High Frequency (UHF) radio waves in the Industrial Scientific Medical (ISM) band from 2.400 to 2.485 GHz [50]. Managed by the Bluetooth Special Interest Group (SIG), it is widely used in IoT applications due to its low power consumption. As with WiFi positioning, the BLE system usually uses beacons as both transmitters and receivers, while positioning uses RSS values as measurements. The algorithm categories are: proximity [51–53], multilateration [5,52, 54], fingerprinting [5,52,55], and the integration of multilateration and fingerprinting [5]. The BLE beacons have the advantages of being small, lightweight, low cost, low dissipation, and they are widely supported by smart devices. Compared to WiFi, BLE is less energy-hungry, has greater smart-device support, is more flexible and is easier to deploy. However, it has similar limitations to WiFi, such as high labor costs in fingerprint database building, and performance degrades due to multipath, signal blockages, and RSS fluctuation [5]. Furthermore, since BLE and WiFi share the 2.4 GHz band, BLE systems are frequently disrupted by WiFi and other BLE devices. The literature shows that BLE/INS integrated navigation is mostly realized by RSS [14,56–58] and sometimes by ToA [59]. Furthermore, since Bluetooth 5.1 protocols were released in 2019, its Angle of Arrival (AOA) positioning accuracy has improved [60,61].

2.1.6. Radio frequency identification

Another popular technology for many IoT applications, RFID often employs Radio Frequency (RF) RSS and AOA to indicate the distance between transmitter and receiver. Its tags, such as active, passive, and semi-active, can be selected for positioning, and its accuracy varies from centimeter-level to room-level, depending on the distribution of tags and the selection of positioning algorithms [6]. The desirable features that have made it so attractive, include contactless communications, high data rate, high security, non-line-of-sight readability, compactness, and low cost [62]. In this positioning method, the RFID reader is attached to a tracked object while active and passive tags are placed on the ceiling and floor [63]. Its potential application scenarios are indoor positioning and pedestrian navigation.

2.1.7. Ultra-wideband

UWB wireless communications, which communicates with a pulse in a very short time interval (>1 ns), and without using a carrier, operates in three categories: communications, positioning, and radar [7]. Its positioning algorithms can be categorized into four groups: TOA [64,65], TDOA [66], AOA [67], and RSS [68]. Its high bandwidth and extremely short pulse waveforms make it effective in Non-Line-Of-Sight (NLOS) environments, reduces the effects of multipath, consumes less energy, and achieves centimeter-level positioning accuracy [47,69]. However, extra infrastructure is needed and signals become disturbed by liquids and metals; it can also be interfered with by other systems operating in the ultra-wide spectrum due to misconfiguration [69]. Also, the extremely short pulses may cause longer synchronization time.

2.1.8. Ultrasonic

Inspired by bats' night navigation, ultrasonic systems, which have been researched for several years, can be categorized into two groups, (i) using a tag as a receiver mounted on the object, and multiple transmitters installed on a wall or ceiling [8]; and (ii) using a tag as a transmitter mounted on the object and multiple receivers installed on a wall or ceiling [70]. Depending on the accuracy required and the operator's budget, ultrasonic positioning mainly adopts trilateration and proximity [8] to process the TOA or TDOA measurements [8,70]. Ultrasonic can achieve centimeter-level accuracy with densely distributed nodes and room-level accuracy with sparsely distributed nodes. This system offers several advantages, including low system cost, high reliability and scalability, high energy efficiency, and most importantly, zero leakage between rooms [71]. However, its performance may vary in different environments because the transmitting speed of the sound in the air can easily be affected by humidity and temperature. Furthermore, its performance may suffer from reflected ultrasound signals and environmental noise [72].

2.1.9. Magnetics

Most applications of low-frequency quasi-static magnetic fields in short-range position and orientation measurements are based on free-space field geometry [73]. They provide a positioning solution with an accuracy of several meters without the aid of other systems and can compensate for some wireless-based positioning technology. Indoor artificial disturbances created by electrical currents in metal or other conducting structures may disturb wireless-based positioning, but magnetic abnormalities created by these interferences can be used as fingerprints or landmarks [74,75]. However, magnetic solutions are not universally applicable, as they can result in significant localization errors.

2.1.10. Odometer

An odometer is an instrument used for measuring the rotation of the wheels of a land vehicle and giving information on the traveled speed and distance. Odometers have traditionally been fitted to the transmission shaft; however, most recent vehicles have an odometer on each wheel. Thus, an odometer is also known as a wheel speed sensor. By differentiating left and right odometer measurements, the yaw rate of the vehicle may be measured, which is a technique known as differential odometry [24]. Please note that the odometer is different from the word 'odometry', of which the latter is a method using data from motion sensors (e.g., camera, LiDAR, odometer) to estimate the change in position over time. Since odometers are commonly used on land vehicles, plenty of works on GNSS/INS integration used the aid of an odometer [10,76,77]. It is useful to improve the accuracy of position, velocity, and attitude during long GNSS outages [76].

2.1.11. Vision

Vision-based systems, using cameras installed on surveillance and smart devices allowing images, or videos, to estimate an object's position, are more robust than wireless-based systems. They can be categorized into two groups, (i) where cameras are placed in a fixed location [11], and (ii) where cameras are placed on mobile devices [78]. In the first group, image-based camera localization can be classified into two categories [79]: with and without known environment. With known environment, a camera's absolute pose (position + orientation) can be determined by features whose types can be points, geometric, semantic, etc. [80], where points are most widely used, called Perspective-n-Point (PnP) problem [79]. These methods in outdoor environment often need prior information given from other sensors (GNSS [81,82] and magnetics [80]). In unknown environment, that is a Simultaneous Localization and Mapping (SLAM, real-time) or Structure from Motion (SfM, post-time) problem [80]. SLAM's typical positioning technologies are Visual Odometry (VO) and Visual-Inertial Odometry (VIO), in which only relative pose can be solved. The second group is also known as visual tracking [83] and can be integrated with motion detection. Yan et al. [11] used an installed camera monitoring a long corridor for visual tracking to aid PDR.

Extensive research has been conducted on visual odometry and mapping. Vision odometry (VO), which is a likely replacement for traditional odometry, collects image data by a camera mounted on the agent and performs pose estimation by associating homologous points in different images. Well-known vision-based algorithms include MonoSLAM [84], ORB-SLAM [85], DSO [86], OV2SLAM [87]. Readers can refer to [88] for a survey of monocular SLAM, and [89] for a survey of visual odometry and mapping approaches with deep learning as the workhorse.

Motion estimation by fusing vision and Inertial Measurement Unit (IMU) enables many applications in robotics. Considering the complementary characteristics of vision and inertial sensors, VIO is a good inertial navigator, exemplified by a legged, or wheeled, robot working in a factory, a field, or indoors. The literature on visual-inertial odometry and mapping is extensive, and notable work includes MSCKF [90], OKVIS [91], VINS-Mono [92], KSWF [93], OpenVINS [94], and ORB-SLAM3 [95]. Readers can refer to [96] for a survey on visual-inertial odometry and [89] a recent survey on it, based on deep learning.

2.1.12. LiDAR

3-D Light Detection and Ranging (LiDAR) sensor, which can provide direct, dense, active, and accurate depth measurements of environments [12], has emerged as an essential sensor for vehicles [97–99], such as self-driving cars and autonomous UAVs. LiDAR collect point cloud data of surrounding objects with a high sampling rate (typically 10 Hz [12,100]); thus it is useful to determine the pose change of the vehicle itself. LiDAR positioning is mostly a relative positioning method and also named LiDAR odometry [12,100]. LOAM [101] is the most classical real-time LiDAR odometry, which sequentially registers extracted edge and planar features to an incrementally built global map. On the basis of LOAM, there are further studies, e.g., LeGO-LOAM [102], Fast-LOAM [103].

With the help of inertial sensors, LiDAR-inertial odometry (LIO) can considerably increase the accuracy and robustness of the LiDAR odometry by compensating for the motion distortion in a LiDAR scan and providing a good initial pose [12]. LINS [100] introduced a tightly-coupled iterative Kalman filter and robocentric formula into the LiDAR pose optimization in the odometry. FAST-LIO2 [12] proposed a new data structure ikd-Tree that supported incremental map updates at every step and efficient inquiries. LIO-SAM [104] firstly formulated LIO odometry as a factor graph, which allowed a multitude of relative and absolute measurements, including loop closures, to be incorporated from different sources as factors into the system. In self-driving applications, GNSS, IMU, LiDAR, camera, and radar can be fused to complete multiple tasks [105,106].

2.1.13. 5G networks

5G (5th Generation Mobile Communication Technology) localization is a technology using cellular networks [107]. Unlike 2G/3G/4G localization with low accuracy, 5G has a potential to achieve a centimeter-level ranging accuracy [20]. Besides, higher signal bandwidths, denser networks, and Multiple-Input–Multiple-Output (MIMO) technologies also significantly increase the ranging accuracy for signal propagation delay based positioning methods like TDOA [107]. The frequencies allocated for the 5G include sub-6 GHz and mm-wave bands, both of which can be used for localization [108]. Since it is not long after 5G was put into use, researches on 5G positioning mostly used simulated results, seldom with real tests [13,108,109]. Decurninge et al. [109] measured Channel State Information (CSI) data using a 32 dual-polarized antenna array at the base station and used extreme learning machines to estimate a single antenna's position. Shamaei and Kassas [108] used a software-defined receiver to extract pseudorange measurements on 600 MHz bands and their ranging error was 1.19 m. Recently, Chen et al. [13] achieved ToA ranging accuracy of 0.5 m in indoor field tests using 5G new radio carrier phase measurements in the sub-6-GHz.

Single positioning/navigation systems are summarized in Table 2.

2.2. Multi-sensor integrated positioning/navigation systems

Section 1 has given the definition of relative and absolute positioning. Based on this definition, this subsection classified multi-sensor integrated systems into: (i) absolute/relative integration, (ii) relative/relative integration, and (iii) absolute/absolute integration systems.

2.2.1. Absolute positioning sources

Absolute positioning, which determines the object's position and attitude in a fixed reference frame, can be categorized by five aspects: (1) radio-based, (2) light-based, (3) audio-based, (4) field-based, and (5) vision-based. Radio-based positioning is a large family that includes, but not limits to, GNSS, WiFi, Bluetooth, RFID, UWB, and 5G. They use man-made emission source to produce electromagnetic waves in radio frequency bands for localization. VLP (refer to Section 2.1.3), using LED lamps as transmitters, and a photodiode, or camera, as the receiver, is the dominantly used light-based positioning. Audio-based positioning technologies, including ultrasonic [8,9] and acoustic [111] in the audible spectrum, are mainly used indoors. Both ultrasonic and acoustic positioning can be integrated with INS [112,113]. Other types of audio-based hybrid systems can be referred to Section 2.1.8. Field-based positioning navigates by matching the database storing field information. The commonly used field-based positioning sources include magnetics, gravity, and map. Magnetic-based positioning, which is mainly used indoors, has been discussed in Section 2.1.9; gravity matching can be used in underwater passive navigation systems to overcome the INS error accumulation and on land with the terrain contour matching [114]; map matching can aid other sensors with reliable location constraints in indoor positioning [115,116] (in Section 6.2) and is also available in urban environments [117]. Vision is capable of both absolute and relative positioning. In addition, Synthetic Aperture Radar (SAR), which uses active microwave imaging radar, is a kind of generalized visual positioning technology [118].

2.2.2. Relative positioning sources

Relative positioning, which determines the object's position and attitude in a moving reference frame or only determines velocity and angular rate, mainly includes (1) inertial sensors and (2) vision. Since INS and vision have been discussed above in detail, we pay more attention on another inertial system, Pedestrian Dead Reckoning (PDR), here. PDR systems use inertial sensors, which are often combined with domain-specific knowledge about walking, to track user movements. Specific to pedestrians, PDR estimates 2D position by accruing

Table 2
Single positioning systems.

Systems	Setup	Positioning methods	Accuracy	Advantages	Drawbacks	Potential integrated sources
Inertial navigation [1,22–25]	Inertial sensors (accelerometers, gyroscopes)	INS/PDR/Motion constraints	Accuracy depends on time	Self-contained, anti-interference	Navigation error accumulates over time	Almost all sources
GNSS [2,26–37]	GNSS satellites, receivers	TOA	From centimeters to meters	Wide coverage in the whole world	Unstable in dense urban canyons and indoor environments	INS, wireless systems and magnetics
VLP [3,38–45]	LED lamps and photodiode or camera	RSS/TOA/ TDOA/AOA	Few centimeters	Not sensitive to severe multipath interference and electromagnetic interference	Signal blockages	Bluetooth and PDR
WiFi [4,23,46–49]	WiFi routers or access points	Trilateration/ fingerprinting/ Kalman filtering	3–10 m	No additional infrastructure, low price, wide coverage	Susceptible to signal interference and blockage	Inertial sensors, magnetics and Bluetooth
Bluetooth [5,14,50–60,110]	BLE beacons	Proximity/ multilateration/ fingerprinting	1–5 m	Low cost, easy to deploy	Performance degrade by multipath effects and signal blockage	Inertial sensors and WiFi
RFID [6,62,63]	RFID tags	RSS	0.1–2 m	High data rate, high security, non-line-of-sight readability, compactness, low cost	Requires heavy infrastructure for accurate positioning	WSN and inertial sensors
UWB [7,47,64–69]	UWB tags	RSS/TOA/TDOA/AOA	Few centimeters	Low energy cost and insensitive to multipath effect	Signal blockage caused by liquid and metallic materials	GNSS and inertial sensors
Ultrasonic [8,70–72]	Ultrasonic tags	Trilatera- tion/proximity	Accuracy depends on the distribution density of infrastructures	Low system cost, reliability, scalability, high energy efficiency, and zero leakage between rooms	Transmission speed depends on environment and the performance may suffer from the environmental noise	RF signals, inertial navigation and magnetometer
Magnetic [73–75]	Magnetometers	Fingerprinting	Several meters	Low cost	Only work in a local area	GNSS, inertial navigation and WiFi
Odometer [10,76,77]	Odometers	Read from wheel rotation	Accuracy depends on time	Self-contained, anti-interference	Only determine forward and heading changes	GNSS and inertial navigation
Vision [11,78–93,96]	Cameras	Feature Detection and Tracking	Accuracy depends on time	Not sensitive to multipath effect and no extra infrastructure needed	Need large computational resources	UWB, GNSS and inertial navigation
LiDAR [12,97–106]	LiDAR sensors	Feature Detection and Tracking	Accuracy depends on time	Precise laser ranging, high time resolution, not sensitive to multipath effect	Large cost, susceptible to the weather	GNSS, inertial, and vision navigation
5G [13,20,107–109]	5G base stations and antennas	RSS/TOA/TDOA/ AOA/CSI	Submeters	High ranging accuracy, high capacity, high reliability	Signal blockages and multipath	GNSS

{distance, heading} vectors representing either steps or strides [119]. Through smartphone market penetration, inexpensive MEMS sensors become ubiquitous and give rise to the wide use of PDR. To prevent PDR from error accumulation, radio-based positioning technologies that are available on smartphones (e.g., Wifi [46,120,121], BLE [122–124], and UWB [125,126]) can be integrated with it.

2.2.3. Absolute/relative integration

Fusion of one absolute and one relative positioning systems is the standard type of the integrated navigation/positioning system. Since most hybrid systems have inertial sensors, we classify this one–one integration type into five categories: (1) radio/inertial, (2) light/inertial, (3) audio/inertial, (4) field/inertial, and (5) vision-based fusion. These inertial systems include INS and PDR. An integration system with relative positioning but without inertial systems are very rare, e.g., GPS/VO [127].

These five categories differ physically, therefore, their usages and advantages are different. Radio/inertial is the dominant integrated navigation system, due to the extensive coverage of GNSS and the wide usage of WiFi and BLE. This category can have wide coverage and system capacity, but its accuracy is susceptible to multipath interference and NLOS, especially indoors. Similar to radio, light-based positioning uses light wave for positioning. VLP suffers more from signal blockages and narrow coverage, but is not sensitive to fading, multipath, and electromagnetic wave interference, and is more accurate in indoor environments [3]. Audio's propagation speed is much lower than that of electromagnetic waves. Thus, it is much easier to measure the Time of Flight (ToF). Audio-based positioning is accurate in ranging but is not robust in the dynamic cases [111]. Different from the three categories above, field-based positioning does not suffer from environmental interference. On the contrary, complex scenes may contain more features for localization (e.g., a capsule endoscope positioning system [128]), which makes field matching an ideal technology to compensate for other absolute positioning systems. Vision-based fusion systems herein do not cover VO and VIO, since vision here is absolute positioning. Its fusion sources include GNSS [81,82], magnetics [80], BLE [129], WiFi [130], and inertial sensors [11,130], which has been discussed in 2.1.11.

2.2.4. Relative/relative integration

Besides the typical absolute/relative integration, there are fusion systems including two relative positioning systems, most of which include inertial sensors. With the low cost of cameras and the development of computer vision, vision is a common choice to aid INS to retard the error accumulation [131–134], and thus, gives rise to Visual-Inertial Odometry (VIO, discussed in Section 2.1.11). Visual-inertial integration is applicable to a wide range of scenes, including indoor [131], outdoor vehicles [133], surface ships [134], and outer space [132]. To further restrain error accumulation, visual-inertial systems can be aided with an absolute positioning source [131,132,134].

Other relative positioning technologies including odometer, speed log, Doppler velocity log (DVL), PDR, and LiDAR can also aid INS, depending on the application scenarios. The odometer and DVL are all instruments that measure speed. Land vehicles are usually equipped with odometers; thus, they are very convenient to be integrated with inertial sensors [10,76,77]. Ships and Autonomous Underwater Vehicles (AUVs) that are not equipped with odometers can use speed logs as an alternative [135–137]. They measure velocities depending on Doppler effects. Pedestrian users can use PDR to integrate with INS [138], although they may share the same inertial sensors. The miniaturization of LiDAR gives it a larger stage for land vehicle [97,98], especially automated driving [99], which will be discussed in Section 6.1.

2.2.5. Absolute/absolute integration

Without relative positioning, there are also works using two absolute positioning sources. Two radio-based systems, although shares similar physical properties, can be integrated together to compensate with each other. These works include WiFi/BLE [52,139,140], RFID/Wireless Sensor Network (WSN) [141], and GNSS/5G [142,143]. Since radio can connect vehicles via communication channels, radio-ranging-based positioning technologies, e.g., Dedicated Short-Range Communication (DSRC) [144] and UWB [145], can be used to measure the vehicle-to-vehicle distances for cooperative localization. But these systems can hardly get velocity and attitude information as inertial sensors do, and may suffer from low sample rates [139].

Due to the shortages of INS-free fusion systems, fusing two absolute positioning sources with an inertial sensor is an alternative. GNSS can hardly provide reliable positioning results in sheltered conditions while other radio-based sources can if infrastructures are provided. In this way, WiFi [33], RFID [146], BLE [59], and UWB [29,35,36] can be integrated with the GNSS/INS system to further improve the performances under GNSS-challenging conditions. Similarly, two radio-based systems excluding GNSS can also be integrated with the inertial sensors (e.g., RFID/UWB/PDR integration [125], WiFi/BLE/INS [14]).

Each absolute positioning category has its own physical properties, thus, multi-categories fusion may better compensate for each other. Radio-based and field-based positioning sources are the most commonly used, most of fusion systems with two absolute positioning categories include them. Radio-based systems are susceptible to signal occlusion while field-based systems do not, thus they can compensate for each other. There are works choosing radio/field integration, including GPS/map [147], GPS/magnetics [148,149], UWB/magnetics [150], Wifi/map [115,151,152], and Wifi/magnetics [75,153,154]. Since the frequency bands of the radio differ much from light's, radio-based positioning aided VLP is applicable [155]. As for radio/vision fusion, Gao et al. [118] proposed an INS/GPS/SAR system where GPS/INS integration provides pseudo ranges and their change rates while SAR provides azimuth and pitch. In addition, the works [9,112] used magnetics/ultrasonic integration indoors where magnetics provides distances and ultrasonic provides attitudes.

3. Analytics-based multi-sensor data fusion for integrated positioning/navigation

3.1. Classification based on algorithms

Estimation is extracting the value of a hidden quantity from indirect, inaccurate, and uncertain measurements, the main goal to minimize the state estimation error, while counteracting uncertainties and perturbations [17]. To differentiate estimation approaches from learning-based ones, they are defined as analytics-based methods. In integrated navigation systems, the most widely used method is the KF, introduced by Rudolf Kalman in 1960 [156], which is suitable for linear systems with Gaussian noises. However, most real-world applications are nonlinear and/or non-Gaussian, which limits KF. Consequently, some variants, including the EKF and the Unscented Kalman Filter (UKF) are applied, which are classified as linearization based filters, sigma point based filters, and Monte Carlo based filters.

In the past decade, the PF has become an effective tool for solving nonlinear analytic problems with a non-Gaussian noise distribution due to increased computational power. More recently, graph optimization has become popular among researchers, since it is globally designed and makes full use of historical information. Fig. 3 shows a general classification of the main algorithms used for state estimation in integrated navigation systems.

3.1.1. Kalman filter

Kalman filter is a recursive analytics-based method, using the estimation of the last state and the observation of the current state to find the optimal estimate of the current state, which is widely used in multi-sensor data fusion for integrated navigation systems. KF assumes that variables and noises are Gaussian distributed, and the system is linear. The KF process and its state is described in two variables:

- $\mathbf{x}_k$ represents the system state at time k.
- $\mathbf{P}_k$ represents the covariance matrix, showing the accuracy of the estimate.

The KF operation consists of two phases: prediction and update. In prediction, the filter uses an estimated previous state to estimate current state.

$$\mathbf{x}_k^- = \mathbf{A}\mathbf{x}_{k-1} + \mathbf{B}\mathbf{u}_k \tag{1}$$

where $\mathbf{x}_k^-$ is the prediction state at time k, $\mathbf{A}$ is the state transition matrix acting on $\mathbf{x}_{k-1}$, $\mathbf{B}$ is the control matrix and $\mathbf{u}_k$ is the control vector. The predicted covariance of $\mathbf{x}_k^-$ is

$$\mathbf{P}_k^- = \mathbf{A}\mathbf{P}_{k-1}\mathbf{A}^T + \mathbf{Q} \tag{2}$$

where $\mathbf{Q}$ is the covariance of noise. During the update phase, the filter optimizes the predicted state using current observations to obtain a more accurate new estimate. Both this estimate and its covariance are obtained with a weighting strategy, whose weight is expressed as Kalman gain $\mathbf{K}_k$.

$$\mathbf{x}_k = \mathbf{x}_k^- + \mathbf{K}_k\left(\mathbf{z}_k - \mathbf{H}\mathbf{x}_k^-\right) \tag{3}$$

$$\mathbf{K}_k = \mathbf{P}_k^-\mathbf{H}^T\left(\mathbf{H}\mathbf{P}_k^-\mathbf{H}^T + \mathbf{R}\right)^{-1} \tag{4}$$

$$\mathbf{P}_k = \left(\mathbf{I} - \mathbf{K}_k\mathbf{H}\right)\mathbf{P}_k^- \tag{5}$$

where $\mathbf{H}$ is the design matrix that maps the state space to the observation space, $\mathbf{z}_k$ is the measurement of time k, and $\mathbf{R}$ is the covariance matrix for measuring noise.

Since Kalman filter can only deal with linear functions, the fusion systems should be linearly designed. In [157], a model combined with strong tracking KF and wavelet neural network for INS error compensation was proposed. Strong Tracking Kalman Filter (STKF) was

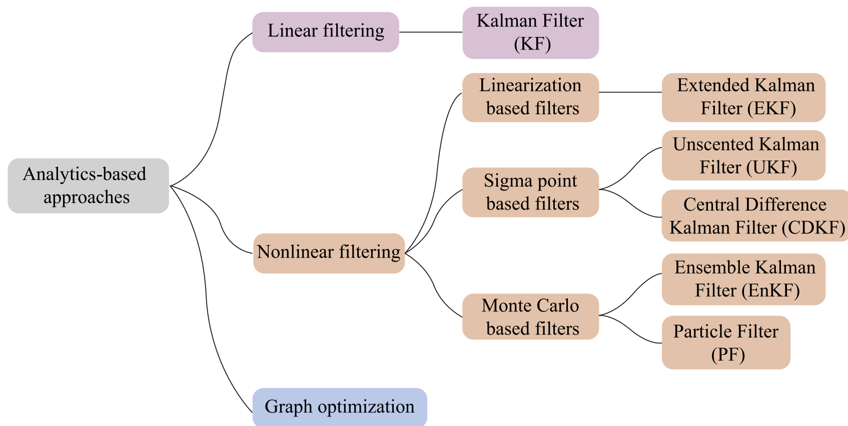

Fig. 3. A general classification of main filters used for state estimation in integrated navigation systems.

used to establish a highly accurate model of when GPS works well and to predict INS errors during GPS outages. KF was applied to integrate INS and UWB in [158–160]. The tightly-coupled model of INS/UWB was established in [159,160]. The work [122] fused the localization results of the Bluetooth propagation model and PDR through the KF to improve the standalone Bluetooth model, step counting, and step length calculation.

3.1.2. Extended Kalman filter

When the system is a linear Gaussian model, the KF can give the optimal estimation, but the actual navigation system always has non-linearity. For nonlinear systems, one suitable method is the EKF, which converts a nonlinear system into an approximate linear system by linearization. The core idea is to expand the nonlinear function into the Taylor series around the filter state estimate and ignore the second-order and above terms to obtain an approximate linearization model. The state transfer matrix **A** and the design matrix **H** in the KF are replaced by the Jacobian matrices of the non-linear functions f and h, which represent the state and observation, respectively..

$$\mathbf{A} = \frac{\partial \mathbf{f}}{\partial \mathbf{X}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} \tag{6}$$

$$\mathbf{H} = \frac{\partial \mathbf{h}}{\partial \mathbf{X}} = \begin{bmatrix} \frac{\partial h_1}{\partial x_1} & \cdots & \frac{\partial h_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_n}{\partial x_1} & \cdots & \frac{\partial h_n}{\partial x_n} \end{bmatrix} \tag{7}$$

With the above matrices, the EKF filtering process is the same as the linear Kalman filtering process.

Many articles use EKF as an efficient algorithm for GNSS/inertial integrated navigation systems [161–164]. To solve the problem of a random bias, the authors of [163] proposed an Adaptive Two-stage EKF (ATEKF). An unknown bias can be estimated using the ratio between the calculated innovation covariance and the innovation covariance estimated from a window of innovations. Xu et al. proposed an adaptive iterated EKF [165] which used the noise statistics estimator to model the measurement noise, and then IEKF was used to deal with the nonlinear problem in the INS/WSN integrated navigation system. A novel approach [166] combined IMU and UWB range measurements by EKF. In [167], the distance between a reference node and the target node and its change rate measured from UWB tags were firstly input to the channel filter for the estimation of the distance. Then, the estimated distance is input to the EKF. In the linearization process, the EKF algorithm may diverge when the neglected higher-order terms in Taylor expansion are significant. The EKF and the PF were used for sensor fusion in pose estimation to minimize uncertainty in robot localization [168].

Algorithm 1 gives an example of GNSS/INS integration to demonstrate how EKF is implemented. The selection of state and measurement vectors will be discussed in Section 3.2. More details concerning INS mechanization and the calculation of matrix **A**, **H**, **Q**, and **R** can be found in [24,169].

Algorithm 1 A GNSS/INS instance [24] showing the implementation of Extended Kalman Filter.

Parameter Statements:

$\{\mathbf{x}_k, \mathbf{P}_k\}$: differential state and its covariance at time step k.

$\mathbf{z}_k$: differential measurement at time step k, using GNSS-observed and INS-predicted quantities.

Initialization:

$\mathbf{x}_0$: zero vector.

$\mathbf{P}_0$: based on the accuracy of initial position, velocity, attitude (PVA), and IMU errors.

Iteration Steps:

(1) INS mechanization:

compute present PVA based on those of last time and IMU outputs.

(2) Prediction:

generate the next state and covariance:

$\mathbf{x}_k^- = \mathbf{A}\mathbf{x}_{k-1}$

$\mathbf{P}_k^- = \mathbf{A}\mathbf{P}_{k-1}\mathbf{A}^T + \mathbf{Q}$

where **A** is the linearization form of the system model,

Q is calculated based on IMU error characteristics.

(3) Update, if GNSS measurements $\mathbf{z}_k$ come:

compute Kalman gain, update state and covariance:

$\mathbf{K}_k = \mathbf{P}_k^-\mathbf{H}^T\left(\mathbf{H}\mathbf{P}_k^-\mathbf{H}^T + \mathbf{R}\right)^{-1}$

$\mathbf{x}_k = \mathbf{x}_k^- + \mathbf{K}_k\left(\mathbf{z}_k - \mathbf{H}\mathbf{x}_k^-\right)$

$\mathbf{P}_k = \left(\mathbf{I} - \mathbf{K}_k\mathbf{H}\right)\mathbf{P}_k^-$

where **H** is the linearization form of the measurement model,

R is calculated based on GNSS error characteristics.

(4) If it is necessary to output:

compute PVA based on INS mechanization results and the state $\mathbf{x}_k$.

3.1.3. Unscented Kalman filter

The UKF belongs to the class of sigma point Kalman filters, which sample a special set of sigma points of the state. Compared to the traditional EKF, UKF has better performance at the expense of computing time because it considers a large number of working points. The most important process of UKF is the unscented transformation where the sigma points are propagated through the nonlinear function and then weighted and resampled into a Gaussian distribution. After the unscented transformation, the sigma points create a vector in the N-dimensional state space. Similar to KF, UKF has two phases, prediction and update. In prediction, sigma points are propagated through the system function for a given control input. For the update step, a new set of sigma points needs to be calculated based on current prediction through the unscented transformation. The detailed algorithm can be referred to in [170,171].

Many papers adopted UKF in an integrated system [170–176]. A variant of UKF, square-root UKF, was implemented in [177–179]. The article [180] proposed a pulsar/CNS integrated navigation method based on federated UKF. The federated filter was used for its stronger fault tolerance and its ability to deal with different sampling periods. For dynamic systems, an Interacting Multiple Module (IMM)-based UKF was proposed [181]. Two IMM-UKFs were executed in parallel when GNSS was available. One fused the information of low-cost GNSS, in-vehicle sensors, and Reduced Inertial Sensor System (RISS), while the other fused only in-vehicle Micro Electromechanical Systems (MEMS) sensors. In [182], a new interacting multiple filter containing simplified UKF-based subfilters with different heading initializations was proposed. It had the same structure as the IMM filter.

Researchers also worked on the modification of UKF. For example, in [183], a computationally efficient refinement of the UKF, called marginalized UKF, was investigated to incorporate the linear substructure in the nonlinear function, i.e., the function was nonlinear only in some of the state elements while the remaining part was linearly mapped to the functional values. Also to reduce the computational burden, in [184], UKF was modified based on the idea that the change of probability distribution for the state variables between measurement updates was small. Another work [185] constructed a nonlinear measurement equation by including the second-order term in the Taylor series of the pseudo-range measurements. To overcome the computational burden, the derivative UKF had the same form as the KF in the prediction step. In the update step, the predicted measurement, measurement covariance, and cross-correlation covariance were approximated by the statistics of chosen sigma points and the transformed sigma points yielded through the nonlinear measurement equation, thus enabling the filtering to retain the nonlinearity of pseudo-range measurements.

3.1.4. Ensemble Kalman filter

The sample points of the previously mentioned sigma-point Kalman filters (i.e., UKF, and Central Difference Kalman Filter (CDKF)) are chosen deterministically. The number of sample points in the sigma-point Kalman filters are of the same order as the dimension of the nonlinear system. However, Ensemble Kalman Filter (EnKF) uses a stochastic sampling algorithm based on the Monte Carlo method, which can better reflect the states of the extremely high order and nonlinear systems. EnKF samples a certain number of points randomly to realize the numerical computation of the state covariance matrix. Such computation not only reduces the computational complexity of calculating the state covariance matrix in KF, but also achieves better performance. Similar to other KF-derived methods, the EnKF also contains a prediction phase and an update phase.

EnKF has also been widely used in integrated navigation systems. In [186], the EnKF was used to estimate the position of the AUV, and the performance of the EnKF with different ensemble members was evaluated. A method of combing EnKF and PF was proposed in [187]. The posterior probability density in EnKF that accounted for the latest observation information was used as the proposal distribution for particles.

Table 3
Comparison Among KF, EKF, UKF, EnKF, and PF.

	KF	EKF	UKF	EnKF	PF
Deal with non-linear functions	Poor	Fair	Good	Good	Good
Deal with non-Gaussian noises	No	No	No	No	Yes
Computational complexity	Low	Low	Medium	Medium	High

3.1.5. Particle filter

The PF is a recursive filter that uses the Monte Carlo method. A set of weighted random samples (i.e., particles) is used to represent the posterior probability of a random event. Different from the KF, which is built on linear state space and Gaussian distributed noises, PF can model any state. Given the particles of the previous state, predicted particles can be obtained from the state transition equation plus the control input. After the observation $\mathbf{z}_k$ is obtained, the conditional probability $p\left(\mathbf{z}_k \mid \mathbf{x}_k^i\right)$, for particles is computed, which represents the probability of observing $\mathbf{z}_k$ when the state $\mathbf{x}(t)$ takes the ith particle $\mathbf{x}^i$. After several recursions, the weights of many particles become small enough to be negligible, leaving only a few particles with large weights. This phenomenon wastes a lot of calculations on particles that have a weak effect. Thus, the estimation performance is degraded. N^{eff} is defined as the number of effective particles. If N^{eff} is less than a certain threshold, re-sampling will slow down the degradation problem. The concept of resampling is to remove particles with small weights and focus on those with larger weights. A typical realization of PF fusion GNSS/INS is given in Algorithm 2. The observation likelihood $p\left(\mathbf{z}_k \mid \mathbf{x}_k^i\right)$ is specified by measurement model $\mathbf{z}_k = h(\mathbf{x}_k, \mathbf{v}_k)$ [10], where $\mathbf{v}_k$ is the measurement noise. The resampling algorithm can be found in [188].

PF is widely used in integrated navigation and has many variants. Woodman and Harle developed a tracking system [189] that used a foot-mounted inertial sensor, a building model, and a PF to track the person and achieved 1 m accuracy. In [190], a Single-hidden Layer Feed-forward Networks (SLFNs) was used to model multiple fingerprinting probabilities and improve the PF performance. A new initialization algorithm using Random Sample Consensus (RANSAC) was presented to reduce the convergence time. In the paper [191], a hybrid extended particle filter was developed as an alternative to the EKF to achieve better navigation accuracy for low-cost MEMS sensors. An adaptive joint observation model was developed to fuse different observations according to the accuracy and reliability of the corresponding sensors. Moreover, an enhanced version of the PF was employed in the paper [10]. It sampled particles from both the prior importance density and the observation likelihood, leading to improved performance.

3.1.6. Summary of filtering methods

A general classification of the above filtering methods can be given in Fig. 3. KF is only capable of dealing with linear functions, so it has limited applications. EKF uses Taylor expansion to linearize the nonlinear functions. Due to the errors introduced by the linearization, the EKF is a sub-optimal and biased estimator, and calculating the Jacobian matrix is always a very difficult and error-prone process. UKF, EnKF, and PF share a similar principle: they use multiple sigma points or particles to fit the nonlinear function without the need of calculating the Jacobian matrix. But their computational cost is much larger. In real applications, the selection of these methods should consider both performance and complexity, which has been summarized in Table 3.

Algorithm 2 A GNSS/INS instance [10] showing the implementation of Particle Filter.

Parameter Statements:

$\{\mathbf{x}_k^i, \omega_k^i\}$: i^{th} particle and its weight at time step k.

$\mathbf{z}_k$: GNSS measurement at time step k.

Initialization:

$\mathbf{x}_0^i$: generated based on initial state's probability distribution $p(\mathbf{x}_0)$.

ω_0^i: $\frac{1}{N}$, where N is the number of particles.

Iteration Steps:

(1) for each particle $\mathbf{x}_{k-1}^i$:

generate the new particle $\mathbf{x}_k^i$ using the system model $\mathbf{x}_k^i = f(\mathbf{x}_{k-1}^i, \mathbf{w}_{k-1}^i)$, where $\mathbf{w}_{k-1}^i$ is the process noise.

(2) for each new particle $\mathbf{x}_k^i$:

compute the new weight using the observation likelihood,

$\omega_k^i = \omega_{k-1}^i p\left(\mathbf{z}_k \mid \mathbf{x}_k^i\right)$.

(3) Normalize the new weight

$$\omega_k^i = \frac{\omega_k^i}{\sum_{j=1}^{N} \omega_k^j}.$$

(4) Resample, if

$$N^{\text{eff}} = \frac{1}{\sum_{i=1}^{N} \left(\omega_k^i\right)^2} < N^{\text{T}}, \text{ where } N^{\text{T}} \text{ is a threshold.}$$

(5) If it is necessary to output:

calculate the weighted mean of particles $\mathbf{x}_k = \sum_{i=1}^{N} \omega_k^i \mathbf{x}_k^i$.

3.1.7. Graph optimization

Apart from filtering-based algorithms, some works adopt nonlinear optimization methods for integrated positioning systems. A graph with state variables as vertices and observation factors as edges is used to describe the inner relationships within the navigation systems through graph-based optimization. Optimization methods were originally used in vision-inertial simultaneous localization and mapping (SLAM) problems [85,86,91,92]. In recent years, they have also been widely used among other positioning/navigation frames, including LiDAR-inertial [104], GNSS/INS [192,193], GNSS/INS/vision [194–196], VLP/INS [197], WiFi/PDR [49], UWB/INS [198], and BLE/PDR [123,124]. Compared with filtering-based algorithms, optimization methods make full use of historical information, thereby achieving better positioning accuracy, although through additional calculations.

Optimization methods are very suitable for sensor fusion, of which the key lies in designing an effective cost function to process multi-sensor data. Generally, a cost function is the sum of the Mahalanobis norm of residuals, which are calculated based on states and measurements. In an integrated navigation/positioning system, each sensor's data can construct at least one type of residuals. For instance, IMU measurements can construct IMU residuals [91,193], PDR residuals [123], or pre-integration residuals [92,195]; image data can construct visual residuals [92,95]. Since batch graph optimization is time-consuming, strategies, such as sliding-window [192,194,197], are needed to reduce the computational cost to realize real-time estimation. In particular, many researches [86,91,92,195] marginalized old measurements outside the sliding window as prior information, which can construct marginalization residuals [92,195]. Also, with effective incremental optimization, GTSAM can conduct real-time localization without a sliding window [193], but the memory consumption is incremental over time. Most researchers choose popular libraries for optimization tasks, including Ceres solver [49,194,197], g2o [123], and GTSAM [192,193].

To estimate the optimal states based on a cost function, effective optimization algorithms are needed. Optimization algorithms can be divided into two major categories, (i) trust region approaches, and (ii) line search approaches [199]. The Levenberg–Marquardt algorithm [200], a trust region approach, is the most widely used for solving graph optimization problems [49,123,197].

We take VLP/INS integration as an example to demonstrate the implementation. The IMU preintegration, IMU measurement residual, and LED projection residual can be referred to [197].

Algorithm 3 A VLP/INS instance [197] showing the implementation of graph optimization.

Parameter Statements:

$\mathcal{X} = [\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_{n-1}]$: the full state vector in a sliding window with a width n.

$\hat{\mathbf{z}}_{b_{k+1}}^{b_k}$: IMU preintegration between the frames at the time step k and $k+1$.

$\hat{\mathbf{u}}_l^{c_k}$: LED l's projection point in the image plane at the time step k.

$\mathbf{P}_l^W$: LED l's prior coordinate in a local reference frame W.

Initialization:

$\mathbf{x}_0$: initial position, velocity, attitude (PVA), and IMU errors.

Iteration Steps:

(1) IMU preintegration:

integrate the IMU measurements within the time interval $[t_{k-1}, t_k]$ to represent the state transition $\hat{\mathbf{z}}_{b_{k+1}}^{b_k}$.

(2) Construct the cost function within a sliding window whose width is n:

$$\min_{\mathcal{X}} \left\{ \sum_{k=0}^{n-1} \left\| \mathbf{r}_{\mathcal{B}}\left(\hat{\mathbf{z}}_{b_{k+1}}^{b_k}, \mathcal{X}\right) \right\|_{\Sigma_{b_{k+1}}^{b_k}}^2 + \sum_{k=0}^{n-1} \sum_{l=0}^{m-1} \left\| \mathbf{r}_{\mathcal{C}}\left(\hat{\mathbf{u}}_l^{c_k}, \mathbf{P}_l^W, \mathcal{X}\right) \right\|_{\Sigma_l^{c_k}}^2 \right\}$$

where $\mathbf{r}_{\mathcal{B}}(\cdot)$ is IMU measurement residual with the covariance matrix $\Sigma_{b_{k+1}}^{b_k}$,

$\mathbf{r}_{\mathcal{C}}(\cdot)$ is LED projection residual with the covariance matrix $\Sigma_l^{c_k}$.

(3) Optimize using the Levenberg–Marquardt method [200] to obtain the optimal state vector $\mathcal{X}$.

(4) If it is necessary to output:

compute PVA based on the state vector $\mathcal{X}$.

3.2. Classification based on integration architecture

Recently, the literature has proposed various types of integration architectures. The level of integration partly depends on whether the architecture is a new system or a retrofit to another system [1]. The three main integration architectures are, (i) loosely-coupled systems, (ii) tightly-coupled systems, and (iii) ultra-tightly-coupled systems. The most popular are GNSS/INS, because they are complementary to each other and they overcome their individual limitations [201].

In integrated systems, filters, such as EKF and PF, are usually used to fuse data from different sensors. In these filters, INS measurements are often used for state prediction; other sensors' observations usually serve as updates to optimize the states. The three types of integration architectures often share similar dynamic equations, but the observation equations are different. A comparison of them are shown in Table 4.

3.2.1. Loosely-coupled integration

The most popular and earliest implemented integration scheme is the loosely-coupled approach, also known as decentralized or cascaded filtering. A Loosely-Coupled Integration (LCI) system uses one system's navigation outputs each time, which often are Position, Velocity, and Time (PVT) rather than raw observations, to update the system states.

We first introduce a general formulation that suits all estimation methods and positioning sources. Similar as the formerly mentioned filtering and optimization-based methods, $\mathbf{x}_k$ represents the system

Table 4
Comparison among three integration architectures.

Architectures	Features	Advantages	Drawbacks
Loosely-coupled Integration	Integrate in the area of PVT, also called decentralized or cascaded filtering.	LCI is a simple and flexible approach.	All the single positioning systems need to independently give solutions.
		It has smaller filter size than the centralized approach.	LCI is less accurate than other schemes.
Tightly-coupled Integration	Integrate in the area of observations, also called centralized filtering.	TCI only needs one system to give independent solutions.	Only one filter is implemented, thus TCI is more complex than LCI.
		It can detect and reject poor observations.	Systematic errors (e.g., cycle slips in GNSS) should be eliminated before filtering.
Ultra-tightly-coupled Integration	Integrate in the physical layer, also centralized.	UTCI has the best performance on signal tracking and anti-jamming capability.	It is not flexible and needs to design when sensors are manufactured. Once made, the architecture can hardly be changed.
		As UTCI can largely improve the receivers' performance, it is theoretically more accurate.	The structure is more complex than LCI and TCI.

state at time k. We assume there were one dead-reckoning-based positioning source and N other types of positioning sources. Their residuals calculated by observations and the estimated state are marked as $\mathbf{r}_{D}(\cdot)$ and $\mathbf{r}_{\mathcal{O}}(\cdot)$, respectively. We use the dead reckoning source as the principle to predict the system state $\hat{\mathbf{x}}_{k|k-1}$ at time k using the sensor's measurements and the last state $\mathbf{x}_{k-1}$. The covariance of the dead reckoning positioning residual Σ_{R_k} and the ith source $\Sigma_{O_k^i}$ are calculated based on the uncertainty of observations and current state. The positioning problem is to minimize the function,

$$\min_{\mathbf{x}_k}\left\{\left\|\mathbf{r}_D\left(\hat{\mathbf{x}}_{k|k-1},\mathbf{x}_k\right)\right\|^2_{\Sigma_{R_k}}+\sum_{i=1}^{N}\left\|\mathbf{r}_{\mathcal{O}}\left(\mathbf{z}_k^i,\mathbf{x}_k\right)\right\|^2_{\Sigma_{O_k^i}}\right\} \tag{8}$$

In the loosely-coupled integration architecture, $\mathbf{z}_k^i$ is one or several components of the state vector and is calculated by the estimator of the positioning source i at time k. To solve this problem, each algorithm has its own optimization criteria: KF, EKF, and graph optimization need to calculate the Jacobian matrix to deal with nonlinear functions, while UKF, EnKF, and PF use particles to fit them.

Specifically, we take the GNSS/INS integration using EKF as an example (Fig. 4), in which the positions and velocities derived from the GNSS estimator, together with their covariance matrices, are employed as updates for the navigation EKF [169]. The construction of transition matrix $\boldsymbol{\Phi}$ can be found in [169]. The state vector of EKF typically contains positions, velocities, attitudes, accelerometer errors, and gyroscope errors, which can be written as:

$$\delta\mathbf{x}=[\delta\mathbf{r},\delta\mathbf{v},\delta\mathbf{A},\delta\boldsymbol{\omega},\delta\mathbf{f}]^T \tag{9}$$

The observation equation of a discrete-time EKF is represented as,

$$\mathbf{z}_k=\mathbf{H}_k\mathbf{x}_k+\mathbf{v}_k \tag{10}$$

For the case using only GNSS-derived positions, they are expressed in Earth-Centered, Earth-Fixed (ECEF) coordinates; therefore, $\mathbf{z}_k$ and $\mathbf{H}_k$ are expressed as,

$$\mathbf{z}_k=\left[L_{\mathrm{GNSS}}-L_{\mathrm{INS}},\lambda_{\mathrm{GNSS}}-\lambda_{\mathrm{INS}},h_{\mathrm{GNSS}}-h_{\mathrm{INS}}\right]^T \tag{11}$$

$$\mathbf{H}_k=\begin{bmatrix}\mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0}\end{bmatrix} \tag{12}$$

For the case considering both GNSS-derived positions and velocities, the observation vector and design matrix are given as,

$$\begin{aligned}\mathbf{z}_k=\big[&L_{\mathrm{GNSS}}-L_{\mathrm{INS}},\lambda_{\mathrm{GNSS}}-\lambda_{\mathrm{INS}},h_{\mathrm{GNSS}}-h_{\mathrm{INS}},\\&v_{\mathrm{E,GNSS}}-v_{\mathrm{E,INS}},v_{\mathrm{N,GNSS}}-v_{\mathrm{N,INS}},v_{\mathrm{U,GNSS}}-v_{\mathrm{U,INS}}\big]^T\end{aligned} \tag{13}$$

$$\mathbf{H}_k=\begin{bmatrix}\mathbf{I} & \mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0}\end{bmatrix} \tag{14}$$

An example on GNSS/INS integration is given in Algorithm 1.

Apart from EKF, other filtering algorithms, such as PF, can also be used [191]. Similarly, Georgy et al. [77] utilized both position and velocity observations for KF and PF in their proposed integration between GPS and RISS. Furthermore, Dusha and Mejias [127] integrated a monocular visual odometry with GPS to improve navigation performance.

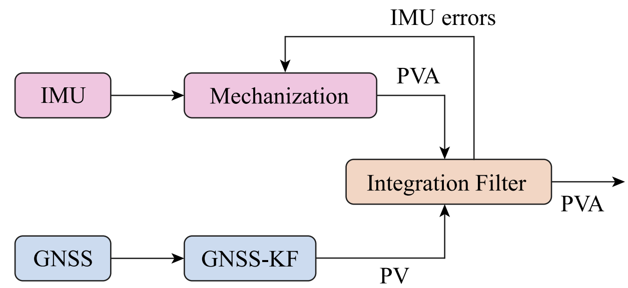

Fig. 4. Loosely-Coupled GNSS/INS integration [169].

Additionally, the article [132] proposed a navigation system that integrates INS, VNS, CNS for deep space exploration. The article [180] designed a Pulsar/CNS integrated navigation method for planetary rovers using UKF. Furthermore, Li et al. [202] integrated miniature coherent altimeter and velocimeter and IMU for Mars landing and Mars based missions in 2010. Besides, Li and Peng [203] proposed the Radio Beacons/IMU integration for Mars entry navigation.

In scenarios without GNSS, loosely-coupled integrated navigation systems are usually implemented through the integration of inertial sensors and other aids, such as UWB [150], magnetometer [204], and WiFi [75,138,205–210]. For instance, WiFi provides several types of observations, including fingerprints [205,209], ToF [208], and propagation model [206,207]. Also, some works employed the integration of velocity-based measurements and inertial sensors. Geng et al. [136] used such integration for AUV navigation. Zhou et al. used the same technique for road driving [211]. While Huh et al. [212] used the Simultaneous Localization and Mapping (SLAM) algorithm to integrate a camera with a gimbaled laser sensor for UAV navigation.

3.2.2. Tightly-coupled integration

Tightly-Coupled Integration (TCI) is also known as the centralized approach [169]. Different from LCI, TCI uses raw observations (including TOA, RSS, AOA, etc.) to update the system states; thus, its number of estimators is fewer [169]. Similar to LCI, TCI also uses the formulation (8) to present the problem. The difference is that $\mathbf{z}_k^i$ is no longer one or several components of the state vector but raw observations of the positioning source i. Thus, the function $\mathbf{r}_{\mathcal{O}}(\cdot)$ is more complicated and needs to be established based on the positioning system's principle.

Specifically, we take the GNSS/INS integration using EKF as an example (Fig. 5), where only one centralized filter is used to integrate the GNSS-derived pseudo ranges, carrier phases, and Doppler frequencies

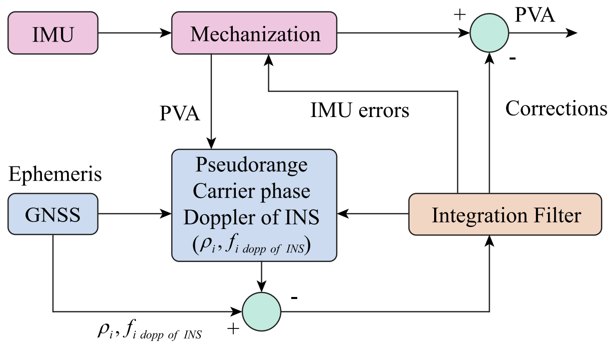

Fig. 5. Tightly-Coupled GNSS/INS integration [169].

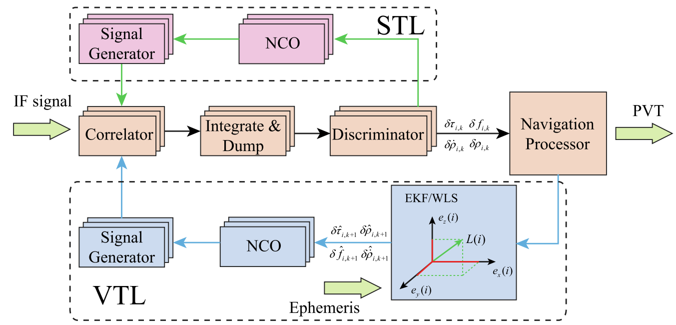

Fig. 6. Block diagram of the STL and VTL of GNSS receiver [224].

with INS-derived Position, Velocity, and Attitude (PVA) [213]. With such a design, GNSS contributes to the integration solution, even when fewer than four satellites are observable.

When designing the tightly-coupled GNSS/INS integrated system, the clock bias and drift errors that affect the GNSS receiver should also be included. The state vector is presented as [214],

$$\delta \mathbf{x} = [\delta \mathbf{r}, \delta \mathbf{v}, \delta \mathbf{A}, \delta \boldsymbol{\omega}, \delta \mathbf{f}, \delta \mathbf{b}]^T \tag{15}$$

where $\delta\mathbf{b}$ usually includes receiver's clock bias δt and clock drift error $\delta t'$, however, some studies only considered the clock drift error $\delta t'$ [215], while some also included the gyro and accelerometer scale factors [169]. For a system without carrier phase measurements, the observation vector $\mathbf{z}_k$ is defined as,

$$\mathbf{z_k} = \varsigma_{\mathrm{sat},k} - \hat{\varsigma}_k \tag{16}$$

where $\varsigma_{\mathrm{sat},k} = \begin{bmatrix}\rho_k & \rho'_k\end{bmatrix}^T$ represents the vector of corrected pseudo ranges $\boldsymbol{\rho}_k$ and pseudo-range rates $\boldsymbol{\rho}'_k$ at the time instant t_k; $\hat{\varsigma}_k = \begin{bmatrix}\rho_k & \rho'_k\end{bmatrix}^T$ represents the vector of predicted pseudo ranges and pseudo range rates, which is computed from the current estimate of the target trajectory [214]. Please note Eqs. (15) and (16) only hold in non-differenced GNSS algorithms, which do not difference measurements as the Real-Time Kinematic (RTK) does.

The observation matrix is time-varying and depends on the number of visible satellites N [214]. The observation matrix $\mathbf{H}_k$ is expressed by,

$$\mathbf{H}_k = \begin{bmatrix} \mathbf{H}_{N\times3} & \mathbf{0}_{N\times3} & \mathbf{0}_{N\times6} & \mathbf{P}_{N\times2} \\ \mathbf{0}_{N\times3} & \mathbf{H}_{N\times3} & \mathbf{0}_{N\times6} & \mathbf{Q}_{N\times2} \end{bmatrix} \tag{17}$$

where $\mathbf{H}_{N\times3}$ is the Jacobian matrix of the nonlinear relationship between the user's position and pseudo ranges $\rho_1,\ldots,\rho_n$. $\mathbf{H}_{N\times3}$, $\mathbf{P}_{N\times2}$, $\mathbf{Q}_{N\times2}$ are given in [214] as follows

$$\mathbf{H}_{N\times3} = \begin{bmatrix} \frac{\check{x}-x_1}{d_1} & \frac{\check{y}-y_1}{d_1} & \frac{\check{z}-z_1}{d_1} \\ \frac{\check{x}-x_2}{d_2} & \frac{\check{y}-y_2}{d_2} & \frac{\check{z}-z_2}{d_2} \\ \vdots & \vdots & \vdots \end{bmatrix}$$
$$\mathbf{P}_{N\times2} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ \vdots & \vdots \end{bmatrix}, \mathbf{Q}_{N\times2} = \begin{bmatrix} 0 & 1 \\ 0 & 1 \\ \vdots & \vdots \end{bmatrix} \tag{18}$$

where d_j is the norm of the vector $\begin{bmatrix}x - x_j & y - y_j & z - z_j\end{bmatrix}$ and $\begin{bmatrix}\check{x} & \check{y} & \check{z}\end{bmatrix}$ is the estimated receiver position. An example on GNSS/INS integration is given in Algorithm 1.

Regarding GNSS carrier phase measurements, it is necessary to establish the relationship between time differenced carrier phase $\boldsymbol{\varphi}_k - \boldsymbol{\varphi}_{k-1}$ and the state vector. The TCI is actually more computationally efficient than the LCI when the ambiguities are fixed [216]. With the development of rapid ambiguity resolution, Li et al. [217] proposed a single-frequency multi-GNSS RTK/MEMS-IMU integration model for positioning in urban environment.

In scenarios without GNSS signals, wireless systems can provide range information for the TCI with inertial sensors, where differences between wireless-based ranges and inertial-based ranges serve as observations. Zhuang and El-Sheimy [23] integrated WiFi, INS, and PDR for pedestrian navigation using an EKF. The observation vector was derived from the difference between WiFi-based and MEMS-based ranging, while RSS bias was also put in the state vector. A similar technique was used for the integration of inertial sensors with RFID [63] and BLE [14]. A UWB system can be tightly integrated with an INS to estimate the indoor position and orientation [218] for indoor pedestrian navigation [167]. Gao et al. [118] integrated SAR tightly with the GPS/INS system to correct INS errors and provide speed-changing information. Also, Gao et al. [98] compared LCI and TCI performance when integrating LiDAR and GNSS with INS.

Alongside range-based technologies, visual-inertial integration systems have been developed recently. Qin et al. [92] designed a monocular Visual-Inertial Navigation System (VINS) for state estimation in various environments, while Visual landmarks and IMU observations were used for optimization. Cui et al. [219] used a tightly-coupled integrated system for Mars exploration, i.e., X-ray pulsars/Doppler system with TOA and velocity measurements. OpenVINS [94] used an EKF-based sliding window visual-inertial estimator for state estimation and camera calibration simultaneously. In recent years, ORB-SLAM3 [95], which fully relied on maximum a posteriori (MAP) estimation, was proposed to fuse monocular, stereo, and RGB-D cameras with inertial sensors.

3.2.3. Ultra-tightly-coupled integration

Ultra-Tightly-Coupled Integration (UTCI), also called deep integration, has been used for applications via GNSS/INS systems, combining GNSS signal tracking and GNSS/INS integration into a single KF. UTCI integrates in the physical layer, rather than the data layer as LCI and TCI do, thus, it is more tightly designed. This method can achieve the best performance for signal tracking and anti-jamming [220,221].

As shown in Fig. 6, the GNSS receiver often has two different architectures: Scalar Tracking Loop (STL) and Vector Tracking Loop (VTL). STL receives Intermediate Frequency (IF) signals, generates errors of pseudo-range, pseudo-range rate, frequency, and carrier phase, and then estimates the Position, Velocity, and Time (PVT) of the receiver. Feedbacks driving the carrier and code Numerically-Controlled Oscillators (NCOs) are obtained from the discriminators directly, while VTL combines two receiver tasks – signal tracking and PVT estimation – into a single algorithm [222]. In vector tracking, receiver feedback is obtained from the navigator to the tracking channels [223], enabling VTL for GNSS/INS deep integration.

Based on the tracking loop algorithm, VTL falls into two groups: coherent [224–226] and non-coherent [149] systems. In a non-coherent system, the discriminator outputs directly determine the code phase and carrier frequency errors. As the carrier phase is not locked [149], the non-coherent system improves performance in weak signal conditions and dynamic environments. Conversely, in a coherent tracking receiver, the Doppler estimate of the carrier loop is fed into the code loop to aid in code tracking as an additional range-rate [227]. A Delay Lock Loop (DLL) employs code tracking and Phase or Frequency Lock Loops (PLL/FLL) to perform carrier phase/frequency tracking [225]. There are also combined Frequency-assisted Phase Locked Loop (FPLL) [223]. Also, a coherent Vector Delay/Frequency Lock Loop (VDFLL) [222,225,228] architecture uses the central filter to track the Pseudo-Random Noise (PRN) code phases and the carrier frequencies, which can be used for dynamic weak signal tracking.

In a typical ultra-tightly-coupled system, the observation updates are directly based on In-Phase (I) and Quadrature-Phase (Q) values generated by correlators. The observation equation is based on the correlation values R_i [221] and expressed as follows.

$$\mathbf{z} = \begin{bmatrix} R_1 \\ R_2 \\ \vdots \\ R_N \end{bmatrix} = \mathbf{H}\begin{bmatrix} \delta x \\ \delta y \\ \delta z \end{bmatrix} + \mathbf{n} \tag{19}$$

$$\mathbf{H} = \begin{bmatrix} -\sqrt{2P_1}D_1R_1''(0) & 0 & 0 & 0 \\ 0 & -\sqrt{2P_2}D_2R_2''(0) & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & -\sqrt{2P_N}D_NR_N''(0) \end{bmatrix}\begin{bmatrix} \frac{\partial \tau_1}{\partial x} & \frac{\partial \tau_1}{\partial y} & \frac{\partial \tau_1}{\partial z} \\ \frac{\partial \tau_2}{\partial x} & \frac{\partial \tau_2}{\partial y} & \frac{\partial \tau_2}{\partial z} \\ \vdots & \vdots & \vdots \\ \frac{\partial \tau_N}{\partial x} & \frac{\partial \tau_N}{\partial y} & \frac{\partial \tau_N}{\partial z} \end{bmatrix} \tag{20}$$

where, P_N is the signal power, D_N is navigation data, τ_N is the propagation delay and $R_N''(0)$ can be calculated by pseudo-random coarse/acquisition code.

Recent research used a Chip Scale Atomic Clock (CSAC) to substitute for Temperature Compensated Crystal Oscillator (TXCO) in the VTL-based deep integration to obtain a precise timing reference, in which only three satellites were needed for navigation [229]. A Vector Tracking Loop Based Deep Integration (VTL-DI) system used vector-form pseudo-range errors and pseudo-range rate errors as observations [229].

Many advanced systems, based on deeply integrated GNSS/INS systems, have been developed. Langer et al. [149] proposed a deep integration for GPS and pedestrian navigation system, which used the step-length update to aid the deeply-coupled GPS/INS system. Fernández [97] used the LiDAR system to help GNSS/INS deep integration. Apart from the GNSS-based deep integration, there was an application of SINS (Strap-down Inertial Navigation System)/CNS [230,231]. He et al. [230] integrated SINS and CNS to determine the rover's position and attitude.

4. Learning-based multi-sensor data fusion for integrated positioning/navigation

This section introduces a general categorization of the learning-based methods (Fig. 7) used in integrated navigation systems. They are generally summarized into two categories, (1) aiding analytical methods to estimate parameters and (2) positioning end-to-end. It is important to note that most novel contributions are based on the combination of several learning based algorithms.

4.1. Supervised learning

Machine learning algorithms are traditionally classified into supervised learning and unsupervised learning, according to whether they require manual data labeling. Supervised learning is a task-driven mechanism focusing on data classification or regression that requires manually labeled data for model training. The most popular supervised learning algorithms used for classification in integrated positioning/navigation systems are SVM and Random Forest. We will discuss several supervised learning algorithms used in integrated positioning/navigation as follows.

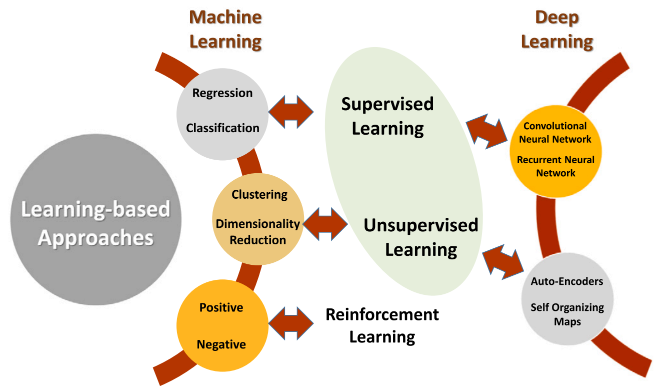

Fig. 7. A general classification of learning-based algorithms is available for most applications. Only the algorithms discussed in this article are included in the classification.

4.1.1. Artificial neural network

The inertial-aided integrated navigation is often affected by uncertain noises [232]. Although UKF and PF have shown their great performance in nonlinear systems, they have the limitations that need the prior probability of the process and measurement noise. Learning-based methods have attracted researchers' interests due to their capability to model and predict in nonlinear systems; hence, many studies have applied neural networks in integrated positioning systems.

In the GNSS/INS integrated systems, ANN can be used to predict positions or INS errors during GNSS outages. In the study [181], two IMM-UKFs were executed in parallel when GNSS was available. One fused the data of low-cost GNSS, in-vehicle sensors, and Reduced Inertial Sensor System (RISS), while the other fused only in-vehicle sensors and MEMS RISS. The differences between the state vectors of the two IMM-UKFs were considered as training data of a Grey Neural Networks (GNN) module to predict and compensate position errors when GNSS signals were blocked. The study [233] mimicked the vehicle dynamics and predicted its position during GPS outages. The method utilized wavelet analysis to analyze and compare the INS and GPS outputs at different resolution levels and then processed the signals by an ANN module based on radial basis function. In [157], a strong tracking KF (STKF) was combined with a wavelet neural network for INS error compensation. The STKF was used to establish a highly accurate model when GPS worked well and to predict INS errors during GPS outages. In [234], Li et al. proposed a single hidden layer feedforward ANN to predict and correct the INS position error during GPS outages. They used the de-noised INS data and the outputs of interactive multi-model EKF to train the learning model.

In addition, ANN is capable of end-to-end integrated navigation. Magrin and Todt [235] fused data from a sonar octagon, a digital compass, and wireless network signal strength measurements to determine the position of an autonomous mobile robot. Almassri et al. [236] designed a neural network to combine UWB and INS data with a motion capture system to guarantee the positioning accuracy of a moving robot.

ANNs have the following advantages. (1) It fuses sensor data for positioning without using the system model, (2) it can predict in complex nonlinear systems without prior knowledge such as characteristics [157], and (3) it is able to deal with massive data efficiently. The challenges of ANNs include: (1) they are black boxes, which cannot explain how each variable affects the navigation system, (2) it usually requires a large number of training data, leading to over-fitting and poor generalization [237], and (3) it is difficult to provide the uncertainty of positioning results.

4.1.2. Fuzzy logic

Standard KF assumes that the process and measurement noises are zero-mean white Gaussian with known covariances. However, in practical systems, it is difficult to model them accurately, which may cause the filter to underperform and diverge. Applying fuzzy logic to implement an adaptive filter is a way to solve these problems, where the system is called Fuzzy Logic Adaptive System (FLAS). In a FLAS, the residual between actual measurements and measurement predictions by the internal model is monitored. Fuzzy logic allows flexibility by using heuristics and avoids the need for accurate knowledge of the process and measurement noise [238]. A typical FLAS consists of three modules: fuzzification, fuzzy reasoning, and defuzzification. The fuzzification module converts a crisp input value into a fuzzy value. Afterward, the fuzzy-reasoning module is responsible for the calculation by using the knowledge base. Finally, the defuzzification module converts the fuzzy actions into a crisp action [239].

Fuzzy logic adaptive KF has been widely used for integrated navigation systems [238–240], where a FLAS was used to adjust measurement noise and process noise covariances of the filter to prevent it from divergence. Authors of [35] calculated the position differences by using the fuzzy neural network. In [170], ultra-tight integration of GNSS and INS was achieved by the fuzzy adaptive strong tracking of UKF. The FLAS was able to tune the softening factor online according to fuzzy rules, leading to performance enhancement in terms of both tracking capability and estimation precision. In [241], the authors proposed a novel fuzzy neural network (FNN) function approximator to model unknown nonlinear systems with real-time implementation for an adaptive fuzzy neuro-observer (AFNO) in low-cost INS/GPS integrated positioning systems.

There are also GNSS-free applications, e.g., for underwater vehicles [242] and UWB/INS fusion systems [243]. Shaukat et al. [242] designed the fuzzy calibration logic to revise the measurement noise covariance matrix for a better fusion of IMU, DVL, compass, and pressure sensor. Gao and Li [243] designed a fuzzy adaption factor to evaluate UWB measurement errors to effectively mitigate NLOS and multipath interferences.

The advantages of fuzzy logic include, (1) it can model process and measurements noise without the need of their probability characteristics [240], (2) the system is simple and justifiable, (3) it takes in expert knowledge and is good at handling nonlinearity and non-Gaussian outliers [242]. The challenges include, (1) its rationale is not constantly accurate, the results are perceived based on assumptions, so it may not be widely accepted, (2) approval and verification of a fuzzy information-based framework need broad equipment-based tests.

4.1.3. Support vector machine

The widely used neural network-based methods demand the quality and quantity of training data, which can be restrictive and limits real-world applications. By contrast, the principle of the SVM is structural risk minimization, rather than empirical error minimization, which is typical for neural networks. SVM also avoids local minima and overfitting in neural networks [237]; hence, it achieves higher performance in certain situations. Authors of [237] explored the Least Squares Support Vector Machine (LS-SVM) to aid GPS/INS integrated systems, especially during GPS outages. The hybrid method consisted of two working modes: the LS-SVM/KF hybrid and the LS-SVM-based prediction during GPS outages. In [172], SVM was trained with the INS data during GPS outage when simulated annealing was applied to optimize the parameters of the kernel and penalty functions. In [137], SVM was used for the navigation of an Autonomous Underwater Vehicle (AUV), which consists of a SINS and DVL. The SVM models the SINS error and makes AUV navigation possible during the long-term DVL outage. Using differently, Song et al. [244] designed two least square SVMs to estimate the distances using RFID inputs and recognize the motion pattern of a vehicle, respectively.

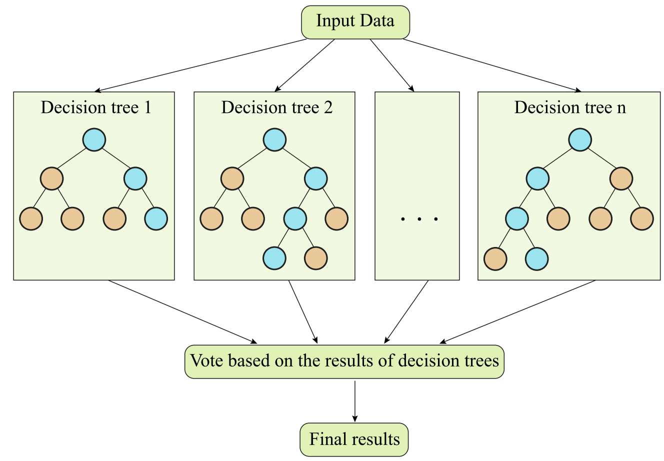

Fig. 8. A typical random forests architecture.

The advantages of SVM include, (1) it has high generalization ability and strong adaptability for different navigation environments [244], (2) it can effectively deal with high-dimensional data, such as wireless fingerprint data, (3) it avoids local minima and overfitting [137,237], and (4) it works well with a small number of training data [137]. The challenges are, (1) time consumption increases obviously on kernel function with a large dataset, (2) its positioning performance decreases significantly if the data has more noise.

4.1.4. Random forest

Random forest is based on decision trees and implemented with three steps, (i) sub-sampling, (ii) decision tree training, and (iii) prediction. In the first step, a fixed number of sub-samples are randomly selected, each of which contains a fixed number of randomly selected features. In the second step, those selected sub-samples are used to train the decision trees. In the last step, all decision trees vote for estimated results. Fig. 8 shows a schematic diagram of the random forest algorithm.

Random forest and its variant, Random Forest Regression (RFR), can both be used for integrated navigation/positioning. Random forest is a classification method, which is be used for human activity recognition [245,246] to fuse PDR with radio-based positioning systems. Besides, the research [247] used the random forest to classify the GNSS positioning accuracy based on GNSS features. RFR is a regression method, which can effectively model the highly nonlinear INS errors due to its generalization capability. In GPS outages, the developed model offers an INS error estimate, thereby improving the continuity and accuracy. Authors of [248] predicted position changes of the target by providing motion information, such as speed and heading angle, into the random forest. The study [249] used a Principal Component Regression (PCR)-RFR hybrid approach to the data containing both linear and nonlinear components, thereby increasing the prediction accuracy.

The advantages of random forest are, (1) it is not sensitive to abnormal data in training, (2) it handles the high dimensional sensor data effectively [248], and (3) it avoids data overfitting [246,248]. The challenges are, (1) in a complex task, many decision trees are needed [245], thus, calculation efficiency is low, and (2) the interpretability of the forest becomes weak when the scale of the forest reaches a certain extent [245].

4.2. Unsupervised learning

Unsupervised learning is a data-driven method and does not rely on manual labeling of data, which uses unknown patterns in features to find the relationships among data samples. In this article, we present k-means [250], Expectation–Maximization (EM) algorithm [251,252], and Principal Component Analysis (PCA) [247]. Since unsupervised learning algorithms are seldom used in integrated navigation systems, we only give them a brief introduction.

K-means is a clustering algorithm to partition observations into k clusters. In the integrated navigation, k-means aids KF to estimate the parameters, such as significant GNSS accuracy-related features [250]. Laskar et al. [253] revised the EKF states by avoiding the clustered obstacles and finding suitable positions.

The EM algorithm is an iterative method of estimating latent variables in statistical models. Huang et al. [252] proposed the EM-EKF algorithm, which combined the EM algorithm with the EKF algorithm, to estimate the positioning information and unknown noise characteristics. The EM algorithm can also be used to combine with UKF [251] and Federated Kalman Filter (FKF) [254] to replace the prediction and update process of KF. Chen et al. [255] proposed an adaptive Student's t-based Kalman filter to integrate GNSS and INS. In these researches, noise is not assumed Gaussian distribution.

PCA is used to reduce the dimensionality of large data sets by transforming a large set of variables into a smaller one that retains the principal information to reduce machine-learning algorithms' parameters. Zhang and Hsu [247] used PCA to identify and discard transient interference in a GNSS/INS integration system. Grejner-Brzezinska et al. [256] used the PCA to reduce ANN training parameters to integrate GPS, IMU, barometer, and other sensors.

4.3. Deep learning

Deep learning belongs to a broad family of machine learning methods based on Artificial Neural Networks, consisting of three or more layers of neural networks. The most representative models of deep learning are the Convolutional Neural Network (CNN) and the Recurrent Neural Network (RNN). CNNs are powerful for dealing with image and point cloud data, therefore they are mainly used in vision [257,258] and laser [259] localization. RNNs allow feedback connections into the same, or previous, layers in order to maintain the memory of past inputs and model problems in time. Since data in integrated navigation/positioning are always sequential, an RNN is an ideal tool.

Unlike other learning-based methods, deep learning is more used for end-to-end integrated navigation problems. In this article, we only explored two papers [257,260] that are used to aid analytical methods. In [257], Siamese CapsNet was proposed for vision positioning, whose results were then fused with UWB for capsule endoscopic localization. In [260], a CNN network was used for walking pattern recognition; and then a regular PDR is adopted, using IMU and magnetometer data.

End-to-end fusion of deep learning can be used in VIO [261–263], PDR [264], vision/magnetic fusion [258], laser/INS fusion [259], and multi-sensor fingerprinting [265–267]. Long Short Term Memory (LSTM), an RNN capable of learning and storing long-term input trends, deals well with time series data and is widely used in VIO [261–263]. VINet [262] used an LSTM to extract IMU data features, which were integrated with image features by another LSTM network. To further improve accuracy, Chen et al. [261] selectively combined image and IMU features, using a two-layer bi-directional LSTM to extract inertial features. By using LSTMs to deal with IMU inputs, Kim et al. [263] designed an unsupervised loss to fuse vision and IMU data. Besides, the work [89] presented a survey of visual odometry and mapping approaches with deep learning. Similar with VIO, Li et al. [259] fused laser and inertial data with a recurrent convolutional neural network composed of a CNN-based point cloud feature extraction, an LSTM-based IMU registration, and an LSTM-based data fusion. In the field of PDR, L-IONet [264], a deep neural networks (DNN) framework, was used to predict poses with the IMU and magnetometer data, which can replace the traditional PDR framework. Deep learning is also capable of fusing multi-sensor fingerprinting data to determine the locations [265–267]. Gan et al. [265] used a deep belief network for multi-sensor fingerprinting, including Wi-Fi, Bluetooth, and geomagnetic data, to obtain more stable and richer fingerprint characteristics. Belmonte-Hernàndez et al. [266] proposed a deep neural network to fuse XBee, Bluetooth, and WiFi fingerprints. Song et al. [267] proposed a hybrid CNN-based indoor localization system (CNNLoc) with WiFi fingerprints for multi-building and multi-floor localization.

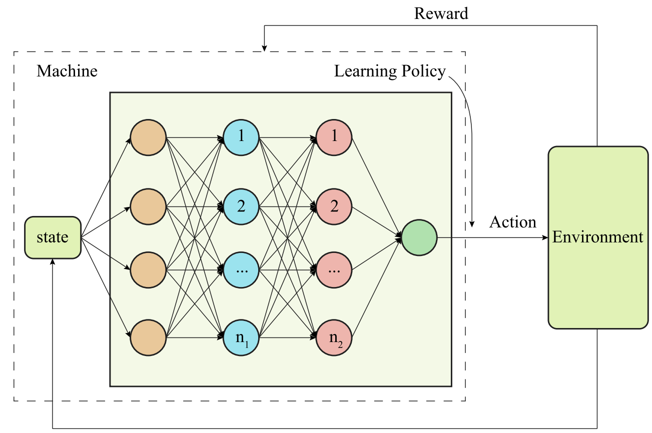

Fig. 9. A typical deep reinforcement learning algorithm architecture.

4.4. Reinforcement learning

Reinforcement Learning (RL) is a basic machine learning paradigm, which, together with both supervised and unsupervised learning, is concerned with exploration of the space of possible control actions to maximize the notion of cumulative reward. Accordingly, a set of actions resulting in good performance were rewarded, while actions resulting in poor performance were penalized [268]. Combined with deep learning, Deep Reinforcement Learning (DRL), as the core of AlphaGo, has been widely studied in recent years. Deep learning provides a learning mechanism, while reinforcement learning offers a learning goal. Fig. 9 is a schematic diagram of the DRL algorithm structure. A neural network decides actions for an Artificial Intelligence (AI) system, while the environment informs whether to reward or punish in order to adjust the parameters of the neural network.

RL has been successfully applied in path planning, collision avoidance [274], and action games, and is now being applied to wireless positioning [269–271] and vehicle tracking [272,273]. Study [269] used RL to adjust the navigation parameters including IMU bias standard deviation. RL provided intelligent solutions by using a combination of dynamic programming and trial-and-error exploration to find a set of optimal parameters. Similarly, Gao et al. [271] adaptively estimated the process noise covariance matrix using a DRL approach, where a deep deterministic policy gradient was introduced to obtain an optimal state. With respect to vehicle tracking, Cao et al. [272] proposed a reinforcement-learning-based Gaussian mixture model to fuse millimeter-wave radars, LiDARs, and cameras in autonomous driving applications. Rangesh and Trivedi [273] presented a modular framework using the reinforcement learning paradigm for multi-object tracking, capable of a variable number of sensors including cameras and LiDARs.

The main usages, advantages, and drawbacks of the learning-based fusion methods surveyed in this paper are summarized in Table 5. The advantages of RL include, (1) it has decision-making ability, which suits well with vehicle tracking, (2) it has a strong automatic exploitation ability, so the learned parameters can keep the validity for a certain period of time [271]. The challenges of RL include, (1) it needs additional storage in navigation applications [271], (2) for DRL, the amount of calculation and the complexity of the reward definition is large.

Table 5
Learning-based multi-sensor data fusion.

Algorithms	Usages	Advantages	Drawbacks
Artificial neural network [157,181,233–236]	Predict positions during GNSS outages.	It fuses sensor data for positioning without using the system model.	ANNs are black boxes, which cannot explain how each variable affects the navigation system.
	End-to-end positioning.	It can predict in complex nonlinear systems without prior knowledge such as characteristics [157].	It usually requires a large number of training data, leading to over-fitting and poor generalization [237].
		It is able to deal with massive data efficiently.	It is difficult to provide the uncertainty of position-ing results.
Fuzzy logic [35,170,238–241]	Adjust measurement and process noise covariance matrices.	It can model process and measurements noise without the need of their probability characteristics [240].	The fuzzy rationale is not constantly accurate. The results are perceived based on assumptions, so it may not be widely accepted.
		The system is simple and justifiable.	Approval and verification of a fuzzy information-based framework need broad equipment-based tests.
		It takes in expert knowledge and is good at handling nonlinearity and non-Gaussian outliers [242].	
Support vector machine [137,172,237,244]	Predict in the fusion systems during sensors' outages.	It has high generalization ability and strong adaptability for different navigation environments [244].	Time consumption increases obviously on kernel function with a large dataset.
		It can effectively deal with high-dimensional data, such as wireless fingerprint data.	Its positioning performance decreases significantly if the data has more noise.
		It avoids local minima and overfitting [137,237].	
		It works well with a small number of training data [137].	
Random forest [245–249]	Recognize human activity.	It is not sensitive to abnormal data in training.	In a complex task, many decision trees are needed [245]
	Evaluate the quality of GNSS positioning.	It handles the high dimensional sensor data effectively [248].	The interpretability of the forest becomes weak when the scale of the forest reaches a certain extent [245].
	Model the high nonlinear INS errors.	It avoids data overfitting [246,248].	
Deep learning [257–267]	End-to-end positioning.	Deep learning is suitable to model data with a large number of inputs including images and time-varying sensor data.	It is a black box, which cannot explain the contribution of each positioning source to the final navigation solution.
	Process camera and laser data.	It can learn complicated features and relationships from sensor data, which is difficult for hand-crafted methods [261].	It requires to train with huge computing resources and massive data.
		With massive training data, the positioning accuracy can significantly surpass other learning-based algorithms.	A large amount of human resources are needed to label the training data.
			It is difficult to provide the uncertainty of position-ing results.
Reinforcement learning [268–274]	Adjust the navigation parameters.	It has decision-making ability, which suits well with vehicle tracking.	It needs additional storage in navigation applications [271].
	Vehicle tracking.	It has a strong automatic exploitation ability, so the learned parameters can keep the validity for a certain period of time [271].	For DRL, the amount of calculation and the complexity of the reward definition is large.

5. Design considerations

5.1. Metrics of navigation/positioning applications

When designing a navigation/positioning application, several metrics should be considered based on users' requirements, which generally include:

(a) accuracy: this is always the most important metric. It mainly depends on the used sensors, which is summarized in Table 2. During the selection of sensors, designers need to consider a tradeoff between accuracy and cost. In addition, accuracy can be evaluated by metrics including mean error and Root Mean Square Error (RMSE) [23].
(b) coverage: the layout of facilities for navigation should be designed based on the required coverage. For the coverage of wireless-based positioning systems, signal coverage and Field Of View (FOV) should also be taken into consideration.
(c) navigation update rate: the requirement for data rate depends partly on users' moving speed. Meanwhile, it is a tradeoff between the update rate and the system burden.
(d) real time consideration: will be discussed in Section 5.4.
(e) scalability: depend on the user scale. It will be discussed in Section 5.5.
(f) availability and reliability: availability indicates the proportion of time when a navigation system can be used, while reliability indicates whether a system can alert users when facing large errors. Both availability and reliability are relevant with the system's robustness to outliers, which will be discussed in Section 5.7.

Apart from these metrics, other software and hardware design considerations are needed to maintain high positioning/navigation performance: analytics-based or learning-based systems (Section 5.2), state selection and observability (Section 5.3), and time synchronization (Section 5.6). Application cases are presented in Section 5.8.

Table 6
Analytics-based versus learning-based systems.

	Pros	Cons
Analytics-based systems	Do not need training data.	System and observation model should be well-studied.
	Needs fewer computational resources.	
	Adaptive to multiple scenarios.	Poorer ability in nonlinear and non-Gaussian systems.
Learning-based systems	Good to deal with non-linear and non-Gaussian systems.	Usually need massive training data and large computational resources.
	Do not need to study the internal mechanism of the system.	Generalization capability is questionable.

5.2. Analytics-based or learning-based systems?

The model-driven analytics-based systems are widely used in industry and have been well tested in practice. However, in this big data era, data-driven learning-based systems become popular and challenge traditional methods for accuracy and robustness; consequently, how to choose between them is a practical question.

Learning-based systems can evolve with more data, which conversely means large computational and data collection costs. However, if not properly trained, learning methods may lead to low generalized ability, i.e., a model trained on one scene may not be transferable [275]. In real applications, the selection between analytics-based and learning-based systems depends on data and computation resources, the maturity of existing positioning models, and the complexity of the positioning task. A comparison of them is summarized in Table 6.

5.3. State selection and observability

In analytics-based navigation systems, the integration system model depends firstly on state selection. This statement is true not only for filtering-based systems, but also for most optimization-based systems [49,192,193]. In this subsection, we mainly discuss the state selection of filtering-based systems, which have been widely used and tested in industry.

Since navigation systems are always nonlinear, it is a common practice to use the error states [24] as expressed in Eqs. (9) and (15), rather than the original states, to describe a system. Most integration systems choose INS or PDR as the main system and design with the error states based on their dynamics. Most integration systems involving INS choose the position and velocity errors in the error states. Attitude errors, accelerometer, and gyroscope biases are usually estimated, the first two can cause a linear growth in velocity error, while gyroscope biases cause quadratic growth. In contrast, most PDR-relevant integration systems choose the position and heading errors in the error states; they seldom choose velocity errors since the position is reckoned by steps, rather than velocities.

Once the state variables are selected, it is essential to take observability into consideration as it can influence the convergence rate in a filtering-based system. The KF measurement model is analogous to a set of equations where the states are the unknown and the measurements are the known quantities. As measurements accumulate over time, there is a set of observation vectors $\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_k$. The linearized measurement model relative to the initial state $\mathbf{x}_1$ is [24]

$$\begin{pmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \\ \vdots \\ \mathbf{z}_k \end{pmatrix} = \mathbf{O}_{1:k}\mathbf{x}_1 + \begin{pmatrix} \mathbf{w}_1 \\ \mathbf{w}_2 \\ \vdots \\ \mathbf{w}_k \end{pmatrix} \tag{21}$$

where $\mathbf{x}_1$ is the state vector and $\mathbf{w}_k$ is the noise vector. The observation matrix $\mathbf{O}_{1:k}$ is defined as

$$\mathbf{O}_{1:k} = \begin{pmatrix} \mathbf{H}_1 \\ \mathbf{H}_2\mathbf{\Phi}_1 \\ \vdots \\ \mathbf{H}_k\mathbf{\Phi}_{k-1}\cdots\mathbf{\Phi}_2\mathbf{\Phi}_1 \end{pmatrix} \tag{22}$$

The rank of matrix $\mathbf{O}_{1:k}$ is defined as m and the length of the state vector $\mathbf{x}_1$ is defined as n. The states are locally observable when $m = n$ and partially observable when $m < n$. The observability of many parameters is dynamically dependent. For example, the attitude errors and accelerometer biases of an INS are both unobservable at constant attitude, but they both become observable after a change in attitude, as this changes the relationship among the states in the system model [24]. Moreover, eigenvalues of the error covariance matrix can be used to evaluate the degree of observability [276]. Observability analysis can also be used to solve the inconsistency problem that real-time estimators, e.g., filters, fixed-lag smoothers, may output falsely optimistic covariance inconsistent to the actual state error [277].

When the state vector is observable, the rate of convergence of a KF state depends on the measurement sampling rate, the measurement noise, and the system noise. This is known as stochastic observability [24]. System and measurement noise can mask the state vector that only has a small impact on the measurements, making those states effectively unobservable. For those stochastically unobservable states, their uncertainties can be equal, or even larger, than the initial ones when the other state variables converge.

In real applications, the states with strong observability should be selected. The choice of inertial instrument errors to estimate depends on their effect on the position, velocity, and attitude solution [24]. An observable IMU error has a significant impact on navigation accuracy and its impact should be no less than that of the random noise. Accelerometer and gyro scale factors and cross-coupling errors are seldom chosen for estimations, since these errors are often unobservable, except in highly dynamic applications [24]. Besides scale factors and cross-coupling errors, the observability of many parameters is dynamically dependent [278,279]. Three instances are given as follows. The attitude errors and accelerometer biases of an INS are not separately observable at constant attitude, but they are after the attitude change [24]. In the vision-aided INS, constant motions (e.g., pure translation and constant acceleration) can lead to additional unobservable directions [278]. In addition, constant accelerometer and gyro measurements can make camera time offset weakly unobservable [279].

The state selection of sensors other than IMU, such as GNSS, WiFi, etc., depends on the integration architecture, as discussed in Section 3.2. In loosely coupled integration, only inertial states are estimated; however, in tightly, or ultra-tightly, coupled integration, parameters of other sensors must be estimated. In GNSS/INS tightly-coupled integration systems, it is common practice to estimate the clock bias and drift errors, as shown in Eq. (15), since they are observable and have a strong impact on the navigation performance. In ultra-tightly-coupled systems, the reference-signal carrier phase offsets $\delta\rho_{i,k}$, for each satellite i and channel k tracked, and carrier frequency tracking errors $\delta f_{i,k}$, etc., may also be estimated [224]. In other tightly coupled integration systems, such as WiFi/MEMS, RSS bias can also be put into the state vector [23]. Furthermore, TOA and its time derivatives can be estimated in a UWB/INS system [218]. Consequently, it is necessary to analyze the system observability, architecture, and application scenario before composing the state.

5.4. Real time considerations

Before designing the positioning or navigation systems, real time consideration is an essential factor that depends on application scenarios. For applications such as car navigation and logistics delivery, real time operation is a hard demand. In applications such as surveying and geo-referencing, a navigation solution is required for analysis offline. Generally, real-time requirements will lead to lower robustness and accuracy.

In analytics-based systems, real time processing is often realized by filtering. Some optimization-based approaches can also offer real-time solutions with a sliding window [192,197] structure. For post-processing systems, optimization methods can achieve a global optimal. Alternatively, Kalman filters can be transformed to Kalman smoothers, of which there are three kinds, (i) fixed-interval, (ii) fixed-point, and (iii) fixed-lag [280,281].

Learning-based systems are also capable of real-time navigation/positioning. Most traditional learning methods, other than deep learning, are hybrids where KF-relevant methods predict the states [35,233,237,249], estimate noise covariances [157,238,239], and other parameters [269]. Since KF is a real-time estimator, most of these methods navigate in real time. Deep learning algorithms are usually designed independently with KF and require high computing resources. With a strong processing unit, they can also achieve real-time positioning in vision-based [258], point cloud based [259], and fingerprint [266] data fusion systems.

5.5. Scalability

Scalability is the metric whether a localization technique can be applied to a large number of users over a large geographical area. Since GNSS is based on broadcasting navigation messages globally and has taken the dominant role in outdoor navigation, it is easy to achieve high scalability in outdoor navigation systems using GNSS technologies, such as network RTK. However, in an indoor environment, positioning infrastructure must be deployed in advance, which may lead to high cost and poorer scalability. A system with good scalability should require less *a priori* information including calibration [282,283], less infrastructure, and lower cost, thus, there is always a trade-off between scalability and accuracy. Infrastructure dependent technologies for high-accuracy indoor positioning is usually more costly; hence, when considering large-scale deployment, it is better to choose technology with less calibration and infrastructure modification. For example, in a fingerprinting system, reducing the offline calibration process by utilizing crowd-sourced data, is an option [283].

5.6. Time synchronization

Multi-sensor data should be timestamped consistently, otherwise, they may interfere with each other. Time synchronization is mostly an engineering problem and is especially important for high-precision positioning, or highly dynamic scenarios [24]. The selection of a synchronized approach depends on whether the sensors can be triggered with an external time Ref. [284]. Some sensors, such as a GNSS receiver, can provide Coordinated Universal Time (UTC) timestamps as a reference, while others, such as radars, are free-running and cannot give absolute time references; in such cases, a sensor should give times offsets relative to other sensors [284,285]. For instance, Lei et al. [285] use UWB timestamps to integrate with IMU and camera. GNSS is an important source of accurate timing and includes UTC timestamps. Hence, GNSS-aided integrated navigation/positioning systems usually suffer less from time synchronization problems.

To date, most studies of time synchronization involve GNSS/INS integration, which can be classified into [286,287], (i) hard (hardware-based), (ii) soft (software-based), and (iii) a hybrid of both. In the hardware approach, the one pulse-per-second signal (1PPS) output from a GNSS receiver is used as input, requiring a specific triggering design, which is unlikely suitable to alternative sensor combinations. The software approach falls into two further categories, (i) the augmented Kalman filter method [287], and (ii) the controlled trajectory method [288]. The performances of both these implicitly relate to state observability and strongly depend on the vehicle trajectory. The hybrid approach utilizes time-tagging by both the hardware and software approaches, which are low-cost, flexible, and reliable, but more involved [286].

Apart from GNSS, clocks inside a PC or another sensor can also be used to generate triggers for synchronization. Jeong et al. [289] used the system clocks of the three PCs to synchronize LiDAR, camera, and GPS/INS systems. Choi et al. [290] used the pulse generated by the clock inside a thermal camera to synchronize other cameras while other sensors (LiDAR and GPS/IMU) are synchronized by software. Similar methods may also be used in GNSS-denied environments.

5.7. Robustness to outliers

Multi-sensor integrated systems are more prone to errors than a single-sensor system. If the unexpected data, such as un-modeled dynamics or data distribution, are not correctly recognized, they will lead to a reduce in overall positioning accuracy, and may even cause the navigation system to diverge. In a robust navigation system, outliers should either be restrained or removed. The most well-known outlier detection algorithm is the RANdom SAmple Consensus (RANSAC). In [291], mismatched points in feature extraction of vision positioning were removed based on the RANSAC method. Some learning approaches, such as the EM algorithm, can also be effective [292]. The methods discussed above deal with outliers in the pre-processing. It is also capable of restraining or removing outliers in the fusion process. In optimization-based approaches, a loss function can be used to reduce the influence of outliers [199]. In filtering-based approaches, when outliers are detected, a robust KF can reduce the Kalman gain and modify the measurement noise covariance matrix [293,294].

5.8. Application cases

In this subsection, we take autonomous driving for instance [105,106,295–297] to analyze the design considerations mentioned above. To safely navigate on the road, centimeter-level accuracy with high reliability is required [295]. Autonomous driving always needs wide coverage on road; thus, positioning systems that support global coverage (e.g., GNSS, INS, vision, and LiDAR) are adopted. These positioning systems also allow good scalability. The navigation system must be strictly real-time and needs a high update rate (>10 Hz [105,295]) since autonomous driving is very sensitive to time. To achieve this, accurate time synchronization (10-ms level [289,290]) is required. Availability, reliability, and robustness are all essential, which means the navigation system should deal well with outliers [296,298]. The navigation algorithm can use the combination of both analytic and learning based methods, e.g., deep learning [105,106,295] is used to deal with vision data while filtering [296] and optimization [297,298] methods are used for state estimation in a fusion system. Additionally, position, velocity, and attitude are all essential when selecting a state.

6. Application scenarios

Multi-sensor integration has been applied for diverse scenarios. The following is a review of typical applications that are classified by navigation platforms. These applications include land vehicles, indoor localization, air vehicles, surface ships, underwater vessels, and outer space, which are presented in Fig. 10.

6.1. Land vehicles

Land vehicles, including cars, motorcycles, vans, mobile robots, etc., have great demands on localization and navigation. In this subsection, we select car and mobile robots as representatives.

Car navigation, which is the most common navigation application, has attracted many researchers. Currently, GNSS/INS integrated navigation [28,30,157,181,182,191,237,239,271] is the most popular

Fig. 10. Application scenarios.

positioning system. In some environments (e.g., urban areas) where GNSS suffers much from multipath, odometers [10,76,77], vision [194,272,299], LiDAR [97,98,296], ultrasonics [300], map matching [117], and UWB [243,301] can be used to aid it to produce a precise and robust navigation system. In recent years, LiDAR and vision become hot in car navigation.

Recently, autonomous driving has attracted huge industry investment. Different from car navigation systems, its tasks include not only localization, but also perception, path planning, and control [295], where localization plays a key role in other tasks. Since autonomous driving is a multi-task process, there are three primary approaches for combining sensory data based on interdependence between the following [106]: high-level, mid-level, and low-level fusion. Autonomous vehicles, which primarily use the camera, millimeter-wave radar, GNSS, IMU, and LiDAR, mainly fuse data from a combination of camera-LiDAR, camera-radar, or camera-radar-LiDAR [105,106], with the aid of GNSS and IMU. In the work [296], a pre-built map was used for LiDAR localization, and multi-sensor data helped with the ambiguity resolution of GNSS RTK. The system made vehicles fully autonomous in crowded city streets. Shao et al. [297] combined GNSS/INS solution with LiDAR scan matching and visual odometry to optimize the poses; then the pose and the LiDAR scan measurements were utilized to build a two-dimensional (2D) probability map. Besides, the tracking of surrounding vehicles is essential for many tasks (e.g., obstacle avoidance, path planning, and intent recognition) and crucial to truly autonomous driving [273]. Research [272,273] used LiDAR and camera to track multiple objects due to their complementary capabilities of environment perception. Multi-sensor fusion in automated driving was reviewed in [105,106].

Localization and navigation are basic requirements when building an autonomous mobile robot system. In outdoor environments, a combination of GNSS, vision-based, and LiDAR-based systems is widely used, while indoors, vision, RF-based, and acoustic systems are popular. Integrating GNSS, RF-based and acoustic systems for mobile robot localization is well established. Compared to visual sensors, sonar is more credible when the robot enters a dark environment. Ultrasonic signals were integrated with inertial sensors [112,302] and wireless network [165,235] to provide localization for an indoor robot. Furthermore, in an indoor environment, RF signals, such as UWB [158,159,236,303], WiFi [304], and WSN [165], can be added to the localized system.

In the past decade, the rapid development of Simultaneous Localization and Mapping (SLAM) [80] has driven camera-based and LiDAR-based mobile robot navigation in an unknown environment. Plenty of VIO and LIO systems including VINS-Mono [92] and FAST-LIO2 [12] have been established for robotic navigation tasks. In [133], the vision-based navigation method was integrated with a low-cost reduced inertial sensor system by EKF to bridge the navigation gap during GPS outages by using IMUs and wheel speed sensors. Tang et al. [195] presented IC-GVINS for wheeled robots, fully utilizing the precise INS in both the state estimation and visual process. Details on vision and LiDAR have been explored in Section 2.1.

6.2. Indoor localization

For indoor localization, the five main positioning systems are, (i) ultrasonic, (ii) RF, (iii) inertial, (iv) magnetic, and (v) vision-based. However, a single positioning system is inaccurate; therefore, indoors, an integrated system is imperative. Usually, a combination of wireless and inertial sensors is commonly used. In [46,116,120,121,138,153,208,305,306], WiFi/PDR navigation systems were improved by introducing estimation filters. WiFi signals can also be integrated with INS [23,33,151,152,189,205,207,307]. Since most buildings have WiFi APs, localization systems do not need additional infrastructure, but power consumption is high. Bluetooth has lower power consumption than WiFi [110]; hence, in [58,306], Bluetooth plus inertial sensors were proposed. By using mobile phones, neither WiFi nor Bluetooth need additional hardware, but they have low accuracy.

Other combinations used RFID [63,125], UWB [126,160,166,167] and WLAN [115,190] to integrate with either INS or PDR. These kinds of systems require additional positioning hardware. An emerging direction of indoor localization is by introducing Visible Light Positioning into the positioning system [3,308,309]. Positioning accuracy is usually better than 0.1 m, outperforming the systems mentioned above. Also, map information [116,151,152,310] and landmarks [305,310] can be utilized for navigation systems, although their construction takes labor and time. Geomagnetic information was used to aid inertial sensors in [74,120,153,311].

6.3. Air vehicles

Air vehicles, including aircraft, guided missiles, UAVs, High Attitude Platform Station (HAPS), etc., can hardly use non-GNSS radio-based positioning sources and map information as land vehicles do. In this subsection, we select aircraft and UAVs as representatives. When multi-sensor integrated navigation systems are applied, GNSS/INS integration is the most popular, for both aircraft [162,163,174,185,312–316] and UAVs [161,164,173,215,317].

An integrated system comprising a SINS and GNSS has become the main method to derive aircraft trajectory information for airborne earth observation [314]. In [312], a sigma-point filter was applied to GPS/INS integration. Attitude was represented by a three-dimensional vector of generalized Rodrigues parameters, which reduced the size of the covariance matrix. An iterated KF was proposed for INS/GPS integration [162] and was applied to SAR motion compensation. Authors of [174] investigated the in-motion alignment of a SINS/GPS integrated system using the UKF and CDKF. In [316], an INS aided by an aircraft dynamic model was presented to manage the absence of GPS signals. A fast-update Aircraft Dynamic Model (ADM) made it possible to utilize a direct filtering method by employing nonlinear INS dynamics as system equations and a nonlinear ADM as observation equations, which substituted for indirect filtering based on linearized error equations. The work [179] addressed the attitude and position estimation of a small-size unmanned helicopter tethered to a moving platform by using a multi-sensor data fusion algorithm based on a Square-Root UKF. The state prediction was performed using a kinematic process model driven by measurements of the inertial sensors onboard the helicopter, while the subsequent correction used information from additional sensors, such as a magnetometer, a barometric altimeter, a LiDAR altimeter, and magnetic encoders that measured the tether orientation relative to the helicopter.

To better utilize UAVs to save effort, time, and human life with the assigned tasks, the UAV requires an accurate, precise, and robust navigation system. In [164], the typical repeated dynamic patterns of

UAVs offered useful information for estimating the UAV navigation states. Machine learning classifiers were employed to detect the repeated dynamic patterns; then an appropriate constraint/update was performed through EKF to obtain a better estimate of the UAV states. Apart from filters, smoothers have also been used in UAV navigation systems. Online and offline UAV position estimation, together with their velocity and orientation, were presented in [317]. For off-line estimation, a fixed-interval smoothing algorithm was applied. Online estimation was accomplished with a fixed-lag smoothing algorithm. To compensate for the GNSS signal outage in an indoor environment, RF signals could be utilized, such as UWB signals [318,319]. The authors of [131] studied the indoor localization of a small, low-cost UAV by using the UbiSense UWB Real-Time Localization System and low-cost IMU in conjunction with an EKF to improve position accuracy. The camera was another widely used sensor during GNSS signal outages [320]. However, vision-based systems suffer in low-textured or low-illumination areas. In [171], a magnetometer was integrated with an IMU for an accurate heading estimation of an indoor UAV.

6.4. Surface ships

In surface ship navigation systems, the GNSS system is an important component as it is often integrated with other sensor modalities to improve navigation performance, such as in GNSS/INS integrated systems [321,322], a GNSS/Laser Gyro INS integrated system [323], and a GPS/IMU/speed log integrated system [135]. However, GNSS signals suffer from many issues, such as spoofs and interceptions [324]. Therefore, many researchers have investigated alternatives, such as the Star Sensor and Radar. To negate outside interference, Star Sensors can work independently and are at the cutting-edge of CNS [325]. The work [326] proposed an algorithm based on a One-Dimensional Wiener Filter (1DWF) to resolve the restoration of the Blurred Star image under high dynamic conditions, by applying it to an integrated SINS/CNS system. The work [325] used an INS/CNS integrated navigation method based on Particle Swarm Optimization Back Propagation Neural Network (PSOBPNN) to resolve CNS's invalidity in cloudy weather.

Currently, autonomous ship navigation is in great demand, with collision avoidance one of its important responsibilities [327]. To this end, radar systems are often integrated with INS or GNSS [327,328]. Moreover, vision-based navigation provides massive amounts of information, does not need wireless devices, and is now hotly debated in the field of surface ship navigation. A ship navigation system has used Vision/Radar/INS integration to measure the baseline between ship and a UAV [134]. The work [329] used a high sampling rate of vision-based navigation to achieve a Shipboard-Relative GPS/INS/Vision integrated system. By fusing INS and Ultrashort Base Line (USBL) in a tightly coupled scheme and without the aid of GNSS, the work [293] achieved promising results. Since trajectories of surface ships are, approximately, within a plane, research [324–326,330] adopted a two degrees of freedom controller.

6.5. Underwater vessels

GNSS signals cannot be used with underwater vessels; consequently, Doppler sonar, inertial sensors, and bathometers are the most common sensors used in navigation [136,137]. Using Kalman filtering to merge data received from an acoustic positioning system, a bathometer and a DVL, a navigation system [331] was developed and implemented in both simulation and in an actual submarine; while authors [332] discussed Underwater Transponder Positioning (UTP), which required only one transponder due to tight coupling with the INS. In [242], USBL is integrated with IMU, DVL, compass, and pressure sensor, which is designed for use in underwater seabed mapping applications for ocean exploration.

6.6. Outer space

There are two main types of integrated navigation systems in outer space, (i) autonomous satellite, and (ii) spacecraft. For the former, CNS is one of the best, since it is low-cost and passive. A satellite's position is estimated by applying the state estimation method considering the measured celestial body positions. Its position error is independent of time and has a strong anti-interference capability; however, it suffers from the low accuracy of the horizon sensor. Another emerging technology is the X-ray pulsar, which compares the pulse TOA observed by the X-ray sensors at the satellite, and predicted by pulse timing [333]. Like CNS, its anti-interference capability is strong and, because it has only one single pulsar, its navigation state is unobservable. Therefore, in [180], a pulsar/CNS integrated navigation method, based on a federated UKF was proposed, and achieved velocity accuracy greater than 0.1 m/s, improved position accuracy, and decreased both cost and failure risk. In [334], a UKF-based information fusion method was proposed for directly sensing horizon by using an earth sensor, and indirectly sensing horizon by observing starlight atmospheric refraction.

For spacecraft navigation systems, some measurements have been used, including geocentric vector, geocentric distance of earth sensor, and star sensor's outputs [335]. The astronavigation system has been widely used in space missions, including CNS and the Doppler measurement. In [238], KF was adapted to integrate data produced by SINS sensor suites and external aiding, such as Autonomous Navigation System (ANS) for guided vehicles to provide optimal estimates of position and attitude. A filtering algorithm, Iterated Square Root UKF (ISRUKF), and a switch-mode information fusion filter based on ISRUKF and EKF were proposed in [335]. This method can fuse geocentric vector and distance measured by a navigation sensor with starlight angular distance and adapt measurement noise covariance matrices online to approximate the actual system. In [336], vision-based relative navigation of two spacecraft was addressed by using the sparse-grid quadrature filter, which provided the estimates of relative orbit and relative attitude together with the gyro biases.

7. Future research directions

Research to date has explored various types of integrated navigation systems, including both analytics-based and learning-based schemes. However, issues and challenges still exist in this field. Here we give some suggestions for future research.

(a) *Accuracy evaluation and error mitigation:* accuracy evaluation is always difficult for localization, especially in some complex indoor environments, such as retail shops, shop floors and offices, because ground truth is often difficult, or even impossible, to obtain. Crowded environments create too much interference, which makes it hard to evaluate errors through external assistance. Multipath and NLoS problems severely affect the accuracy of wireless-based localization. Future work may focus on the detection and mitigation of these problems in integrated systems. Using multiple sensors can help with the detection of multipath and NLoS [223], which can be realized either by environment sensing, or training the historical dataset.

(b) *Collaborative localization and crowdsource-based positioning:* while millions of users and devices are connected by IoT, massive amounts of data are collected in real time. Hence, collaborative localization and crowdsourcing using data from existing public infrastructures is a trend for low-cost mass-market IoT applications. Under the crowdsourcing framework, location users become also location providers; hence, in a fingerprinting positioning algorithm [154], time and labor to build the database are largely saved.

(c) *Broader use of optimization methods:* analytics-based approaches are mainly KF-relevant estimation methods. However, graph optimization is highly likely to deal with nonlinear problems; also, it is easy to design an optimization architecture for fusing multi-sensor data.

With a sliding window design, it can also support real-time position/navigation. In the future, we anticipate seeing more applications with graph optimization approaches.

(d) *Theoretical aspects of sensor fusion:* for instance, the core goal of both analytics-based and learning-based systems is to minimize errors; however, most algorithms are easy to fall into local minima rather than global optimization. Therefore, how to transfer from a non-convex problem into a convex one remains to be explored, especially for optimization-based approaches.

(e) *INS-free filtering model:* in KF-relevant systems, INS error is commonly used as the state model, which makes INS an indispensable part of an integrated navigation system [132]. The state model, without having to depend on INS, and INS-free integrated navigation, should be explored in future studies. This will help to ensure that navigation systems will still work in the event of accelerometer or gyro failure. For that model, observability (demonstrated in Section 5) should be fully analyzed.

(f) *Comprehensive comparisons between learning-based and analytics-based approaches:* learning-based navigation systems perform better when dealing with calibration errors and faulty sensors. However, most research does not thoroughly differentiate between learning-based and analytics-based methods, in that they always discuss the accuracy, but they neglect other qualities, such as reliability and cost; this make it hard to select the most effective localization scheme. A general comparison between these two methods is presented in Section 5. We urge future researchers to concentrate more on method selection for the various integrated systems. In addition, some learning-based methods, such as RL, should receive more attention.

(g) *Positioning with new sensors:* modern technology has given rise to many new sensors, of which many are capable of positioning in a certain scenario. For instance, in urban areas, an event-based camera can capture moving objects, thereby calculating their kinematic characteristics; the 6G communication, together with high-resolution imaging and sensing, will provide 10 cm level indoor positioning [337].

(h) *Interpretability of learning-based methods:* learning-based methods are black boxes [282] and their mechanisms cannot be easily interpreted, especially for deep learning. Poor interpretability means that navigation users will not trust the robustness of a training model; consequently, an integrated navigation system that is short on robustness, may lead to serious accidents. Therefore, a potential research direction is to investigate how deep neural networks learn when they are used for multi-sensor fusion.

(i) *Data variety for indoor localization:* in the era of big data, research on multi-sensor positioning/navigation should not be limited to the positioning algorithm, but also pay attention to data variety, especially for indoor localization. People spend 80% of their time indoors, which enables us to train huge data resources with deep learning algorithms. For instance, pedestrian-based localization plays an important role in LBS applications. However, existing PDR methods usually considered simple walking patterns (forward walking, backward walking, right lateral walking, left lateral walking, and running) [260]. To obtain more accurate location-based services, more walking patterns, such as jumping, jogging, sprinting, ascending and descending stairs, crouch walking, and stair climbing, we expect will be considered in future work. For pedestrian-based applications, we also hope to include more IoT devices, such as smartwatches, wristbands, and smart earphones, to extend the potential applications of the existing research.

(j) *Integrity monitoring:* a navigation system can occasionally produce output errors much larger than the uncertainty bounds specified for, or indicated by it. This may be due to hardware or software failures, or unusual operating conditions. Integrity monitoring systems detect these faults and protect overall navigation solutions. Integrity, together with accuracy, continuity, and availability, are the most required navigation qualities [24]. However, to our knowledge, most studies of integrity discuss filtering-based GNSS/INS systems. Other navigation systems and algorithms have not been fully studied. To build a secure multi-sensor navigation system, more attention should be paid to integrity monitoring in future work.

(k) *Positioning privacy:* Existing studies usually provide the same level of privacy protection. However, users need different levels of privacy protection for different times, different locations, and different data. For example, the user requires a higher level of privacy protection at the point of interest. On the contrary, the user requires a lower level of privacy protection at unimportant locations. Therefore, a personalized location privacy protection is a research hotspot. Secondly, existing solutions ignore the trust evaluation of privacy-preserving systems, leading to misevaluation of the privacy protection degree provided by a privacy-preserving system. Additionally, it is still a challenge to put forward indicators that can correctly measure privacy protection technology. Thirdly, existing researches are inflexible and lack the ability to adaptively select privacy protection strategies for different scenarios. More importantly, with the maturity of technologies such as edge computing and blockchain, it is still difficult to balance privacy protection and utility when applying the above technologies to privacy protection. Finally, with the development of quantum computer, the traditional privacy protection methods based on cryptography cannot resist the computing power of quantum computer. Therefore, location privacy protection against quantum attacks is a hot research topic in the future.

The next decade will almost certainly witness the continued proliferation of integration architectures with further improvement in cost, integration, and performance.

8. Conclusions

This article presented a wide-ranging survey of multi-sensor integrated navigation/positioning systems, which gives a comprehensive understanding to practitioners and researchers. To thoroughly review integrated systems, we explored the literature and discussed many widely-used single positioning and navigation systems. We then classified integrated navigation systems concerning categories of (1) sources, (2) algorithms and architectures, and (3) scenarios. To go into a deeper study, we classified the integrated navigation/positioning systems into two categories based on integration algorithms, (i) analytics-based fusion, and (ii) learning-based fusion. Analytics-based fusion originated from the well-known Kalman filter, therefore it can be classified further, based either on algorithms or integration levels. There has been very little academic discussion regarding learning-based fusion, therefore, we selected several algorithms to both demonstrate and compare with each other: ANN, fuzzy logic, SVM, random forest, deep learning, and reinforcement learning. We have given a comprehensive tutorial on the consideration of design before implementing an application. We have also given a detailed introduction to the applications of integrated navigation/positioning based on application scenarios. Furthermore, we have suggested future research directions in order to trigger readers' thinking about this crucial subject. Finally, we look forward to seeing more breakthroughs of integrated navigation/positioning, and more valuable applications in areas such as IoT.

CRediT authorship contribution statement

Yuan Zhuang: Conceptualization, Project administration, Writing – review & editing. **Xiao Sun:** Investigation, Writing – original draft, Writing – review & editing. **You Li:** Writing – review & editing. **Jianzhu Huai:** Writing – review & editing. **Luchi Hua:** Investigation, Writing – original draft. **Xiansheng Yang:** Writing – review & editing. **Xiaoxiang Cao:** Writing – review & editing. **Peng Zhang:** Writing – review & editing. **Yue Cao:** Writing – review & editing. **Longning Qi:** Writing – review & editing. **Jun Yang:** Writing – review & editing. **Nashwa El-Bendary:** Writing – review & editing. **Naser El-Sheimy:** Writing – review & editing. **John Thompson:** Writing – review & editing. **Ruizhi Chen:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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